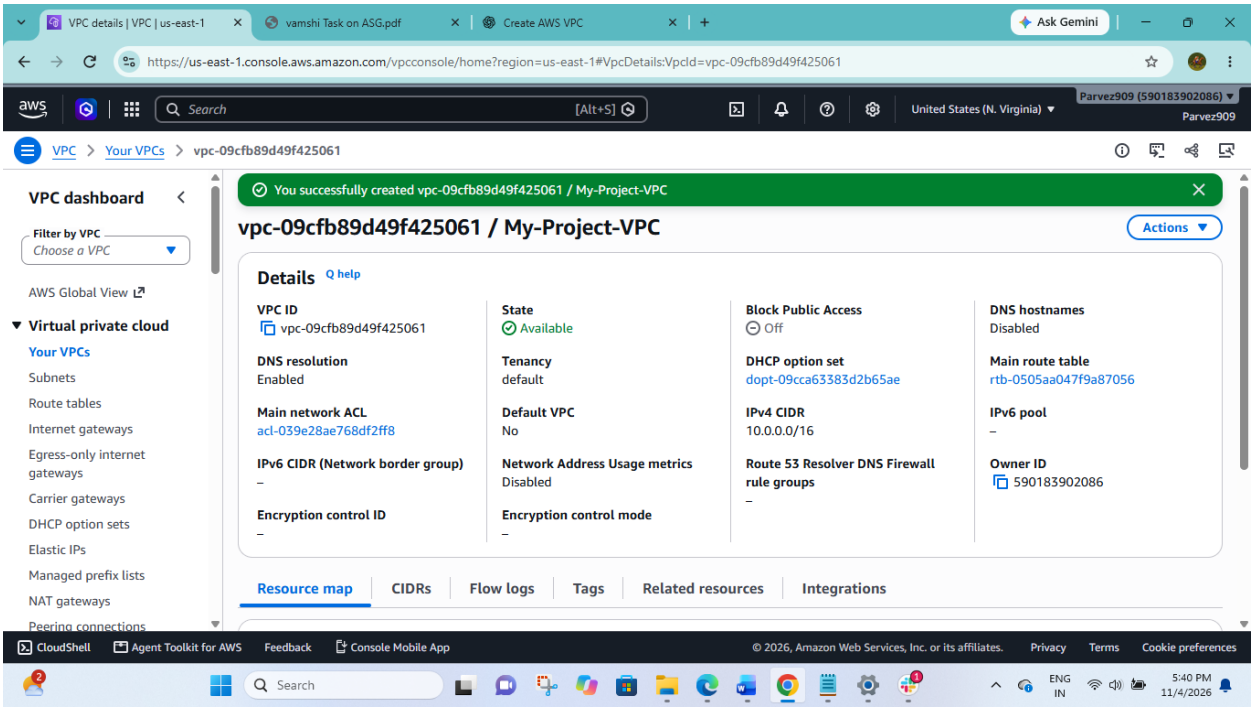
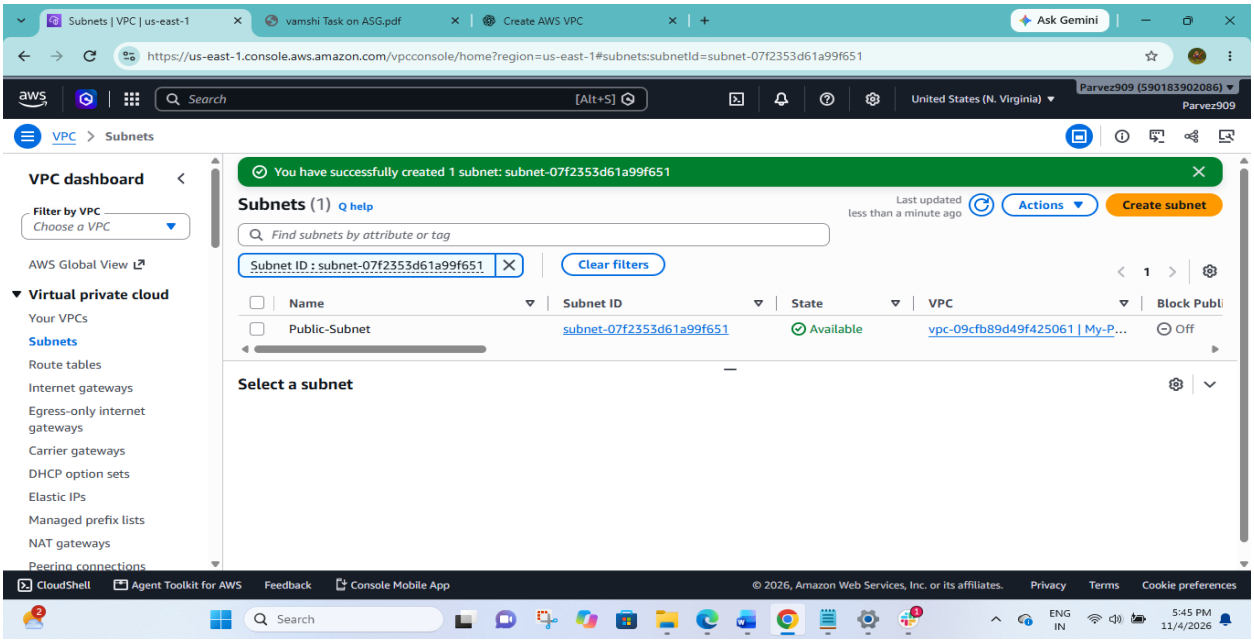
Task on ASG

1. **Create one VPC in N. Virginia region.**

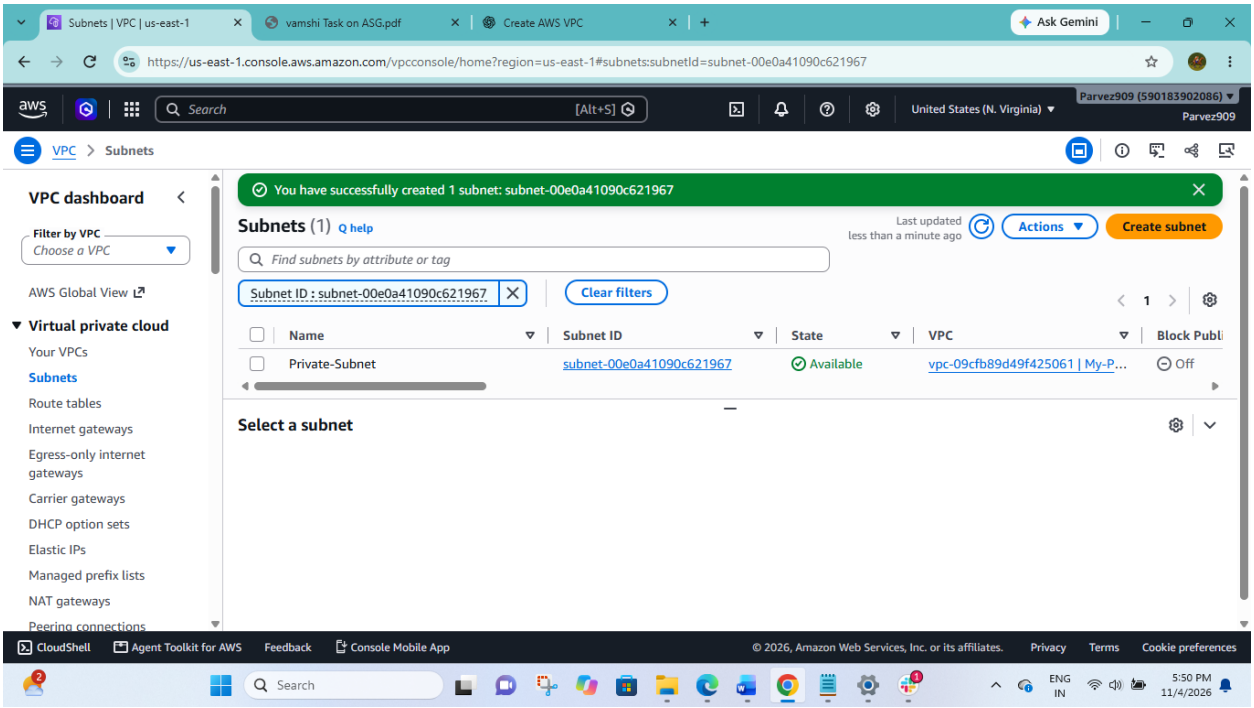


2. **Create two subnets: one public subnet and one private subnet.**

this is our public subnet.



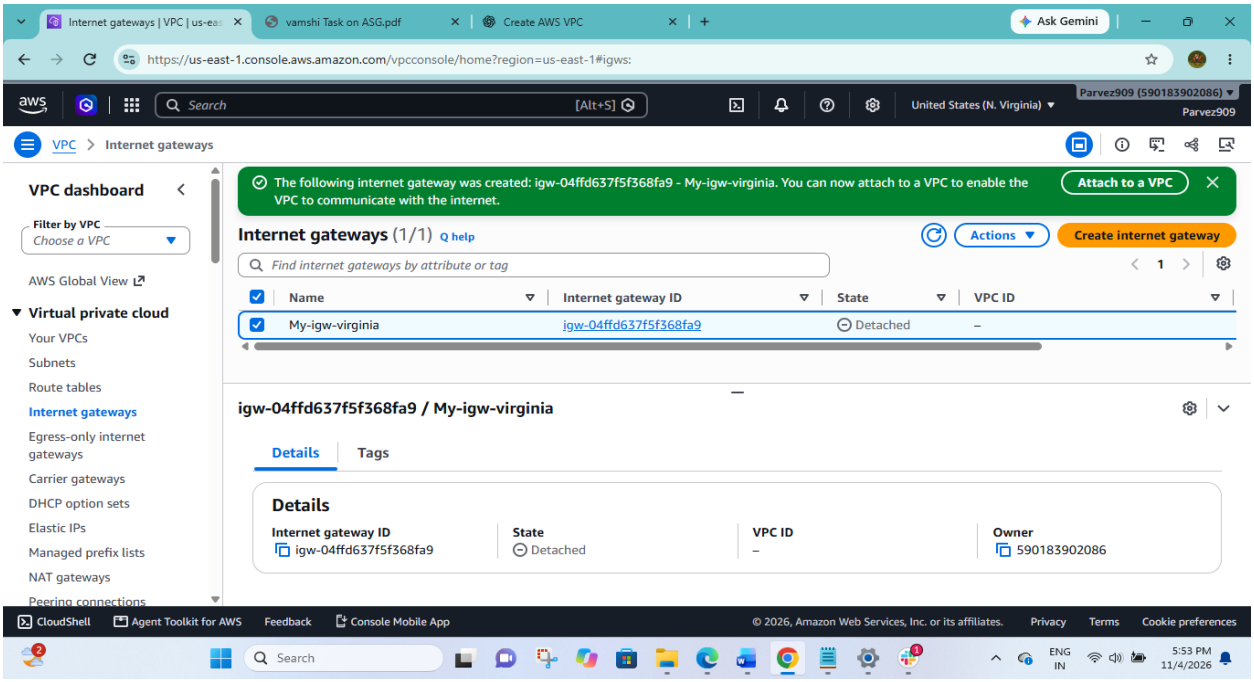
•



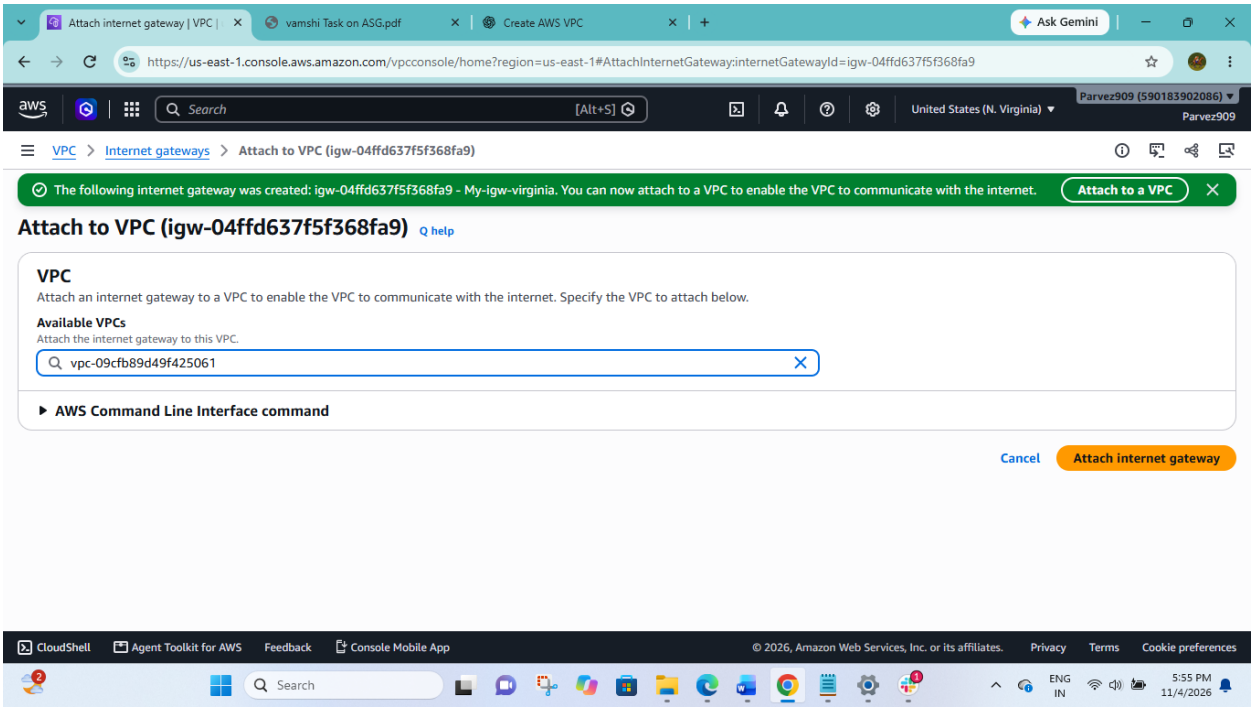
successfully created private subnet.

3. Attach an IGW to the VPC.

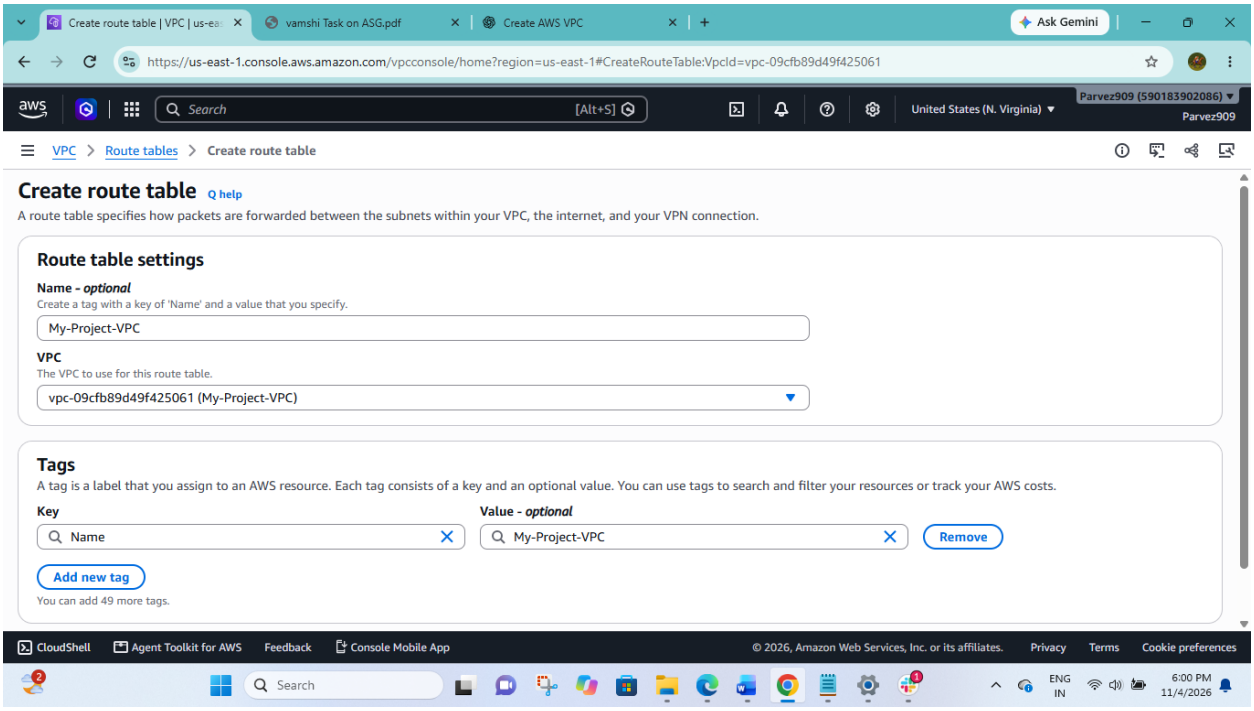
successfully created Igw



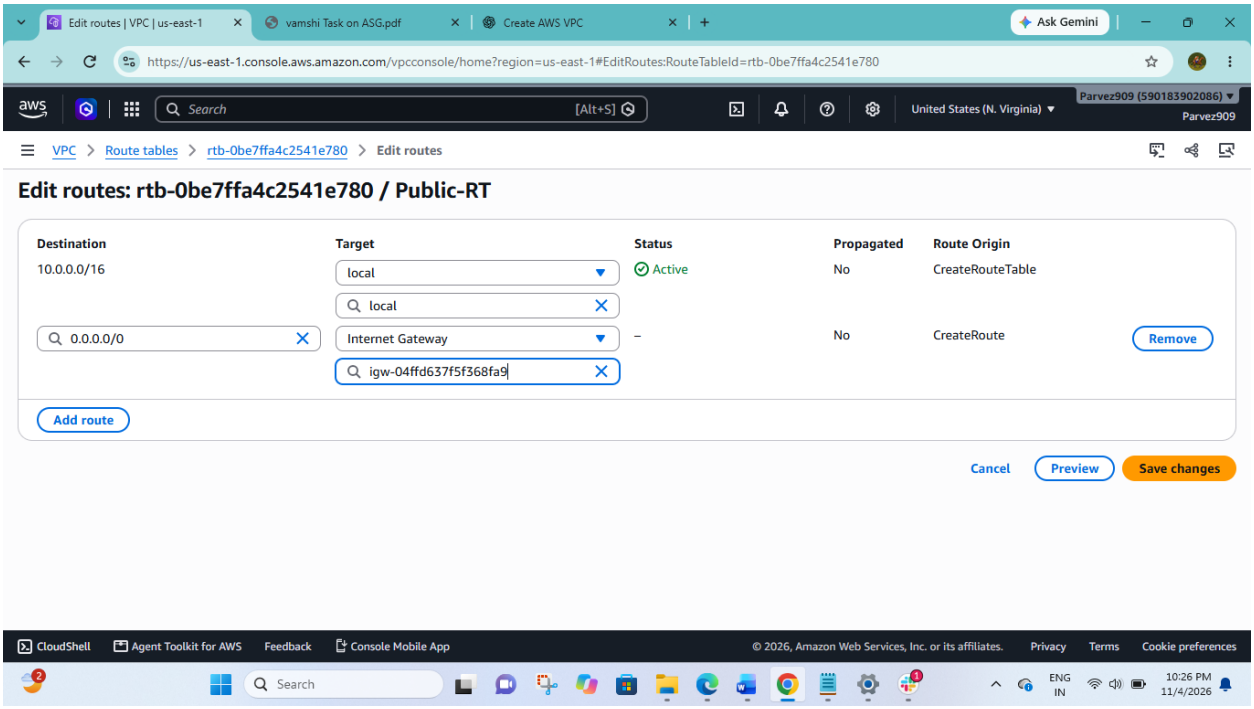
successfully Attached an IGW to the VPC.



4. Create one public route table (RT) and one private route table.



Agg IGW with public-RT



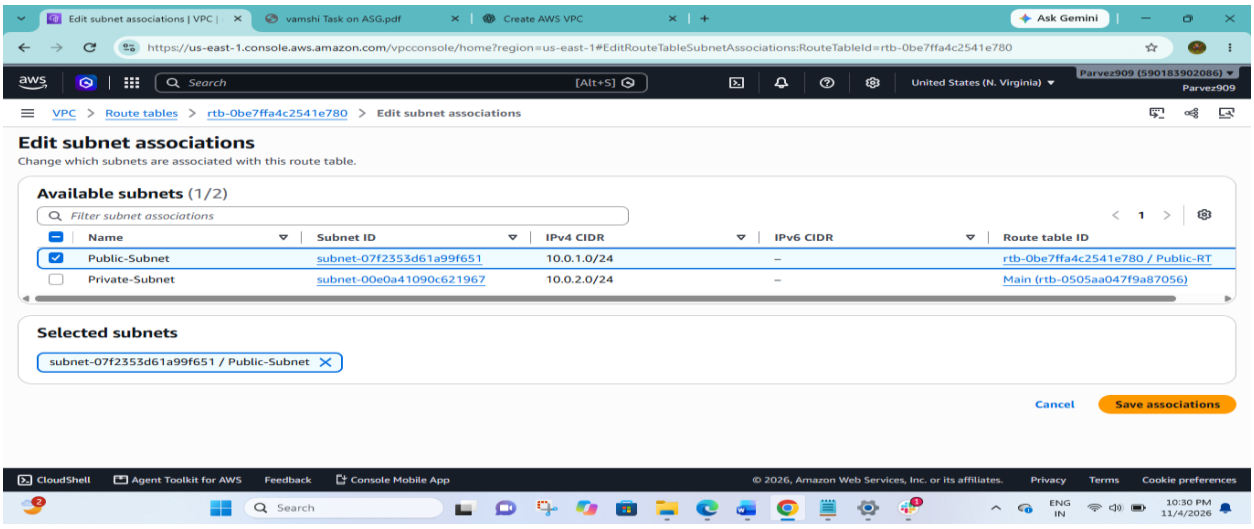
Associate Public Subnet

➡ **Open the Subnet associations tab.**

➡ **Click Edit subnet associations.**

➡ **Select Public-Subnet.**

➡ **Click Save associations.**



Create Private Route Table

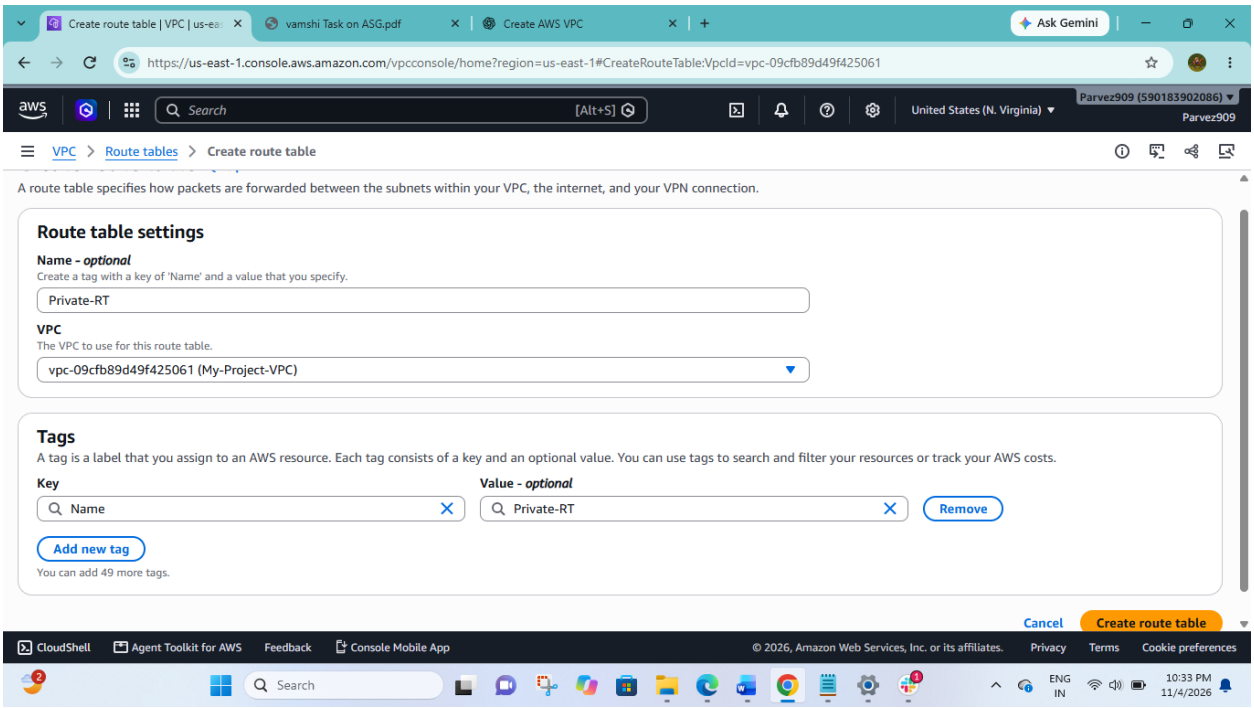
➡️ **Go to VPC → Route Tables.**

➡️ **Click Create route table.**

Name **Private-RT**

VPC **My-Project-VPC**

➡️ **Click Create route table.**



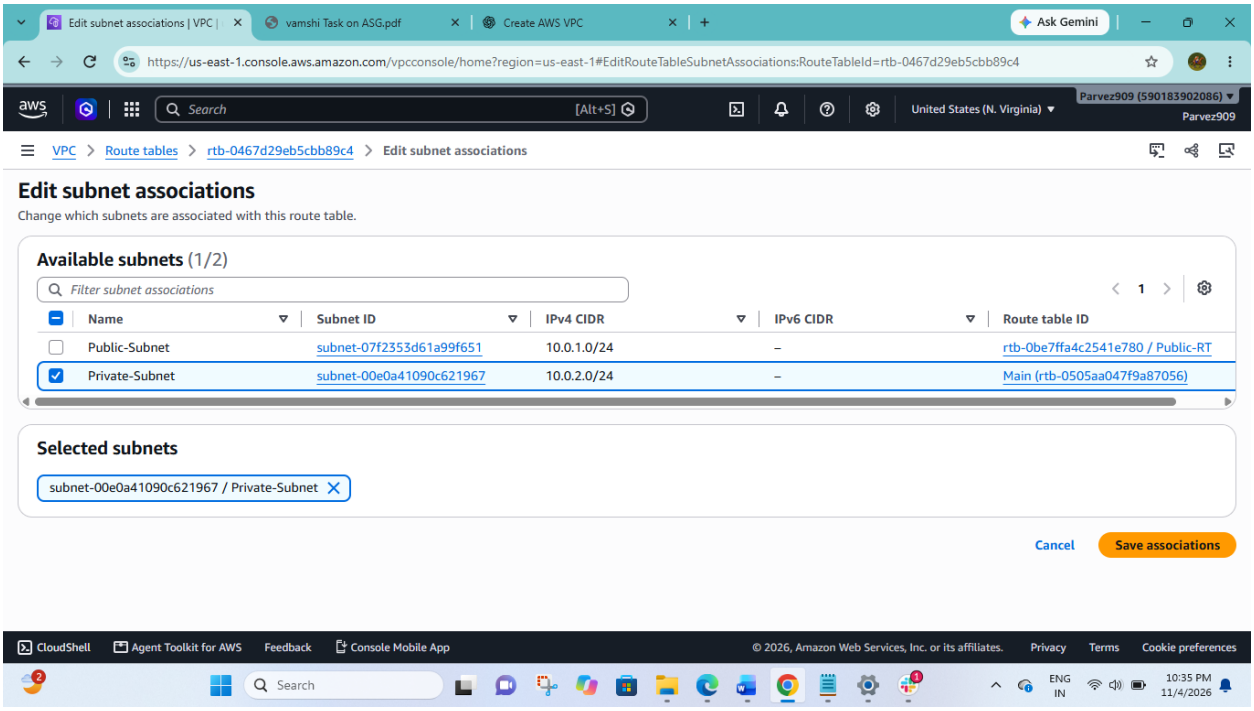
Associate Private Subnet

➡️ **Select Private-RT.**

➡️ **Open Subnet associations.**

➡️ **Click Edit subnet associations.**

➡️ **Select Private-Subnet**



5. **Deploy a NAT gateway in the public subnet and attach the NAT gateway to the private subnet.**

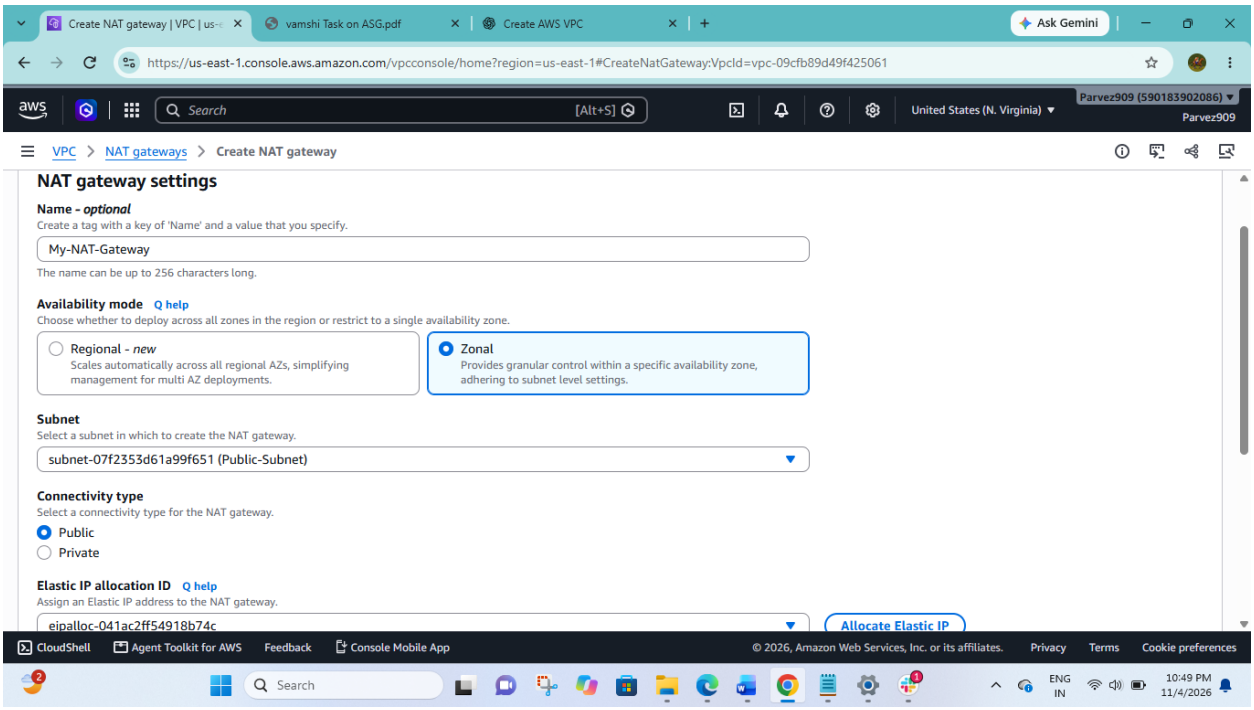
   ➡ **Select Public under Connectivity type.**

   ➡ **Confirm that Public-Subnet is selected.**

   ➡ **Confirm the Elastic IP allocation ID is selected.**

   ➡ **Leave Additional settings and Tags at their defaults.**

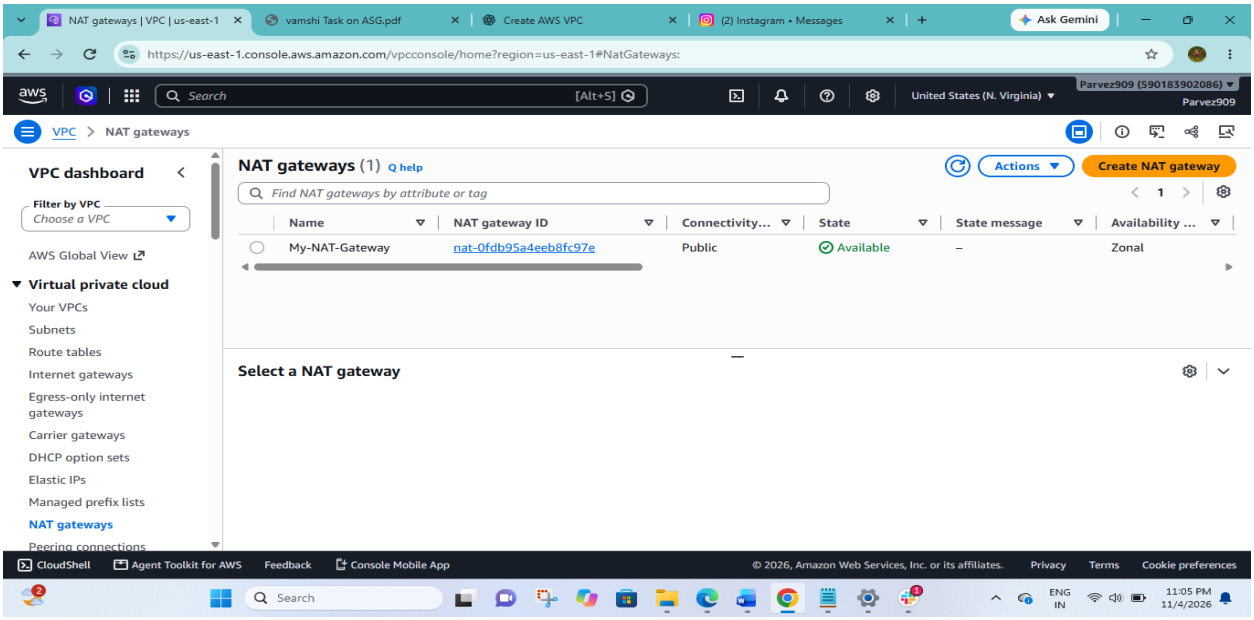
➡️ Click Create NAT Gateway.



After creation

1. **Go to VPC → NAT Gateways.**
2. **Find My-NAT-Gateway.**

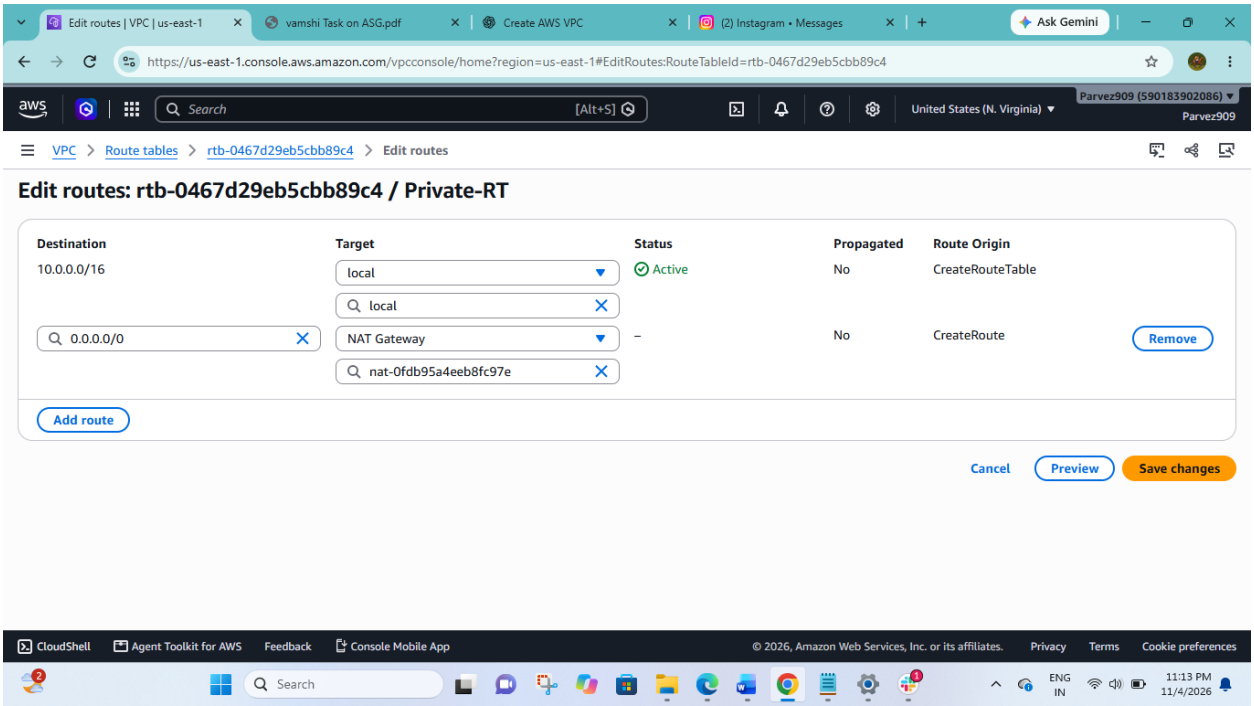
3. Wait until Status becomes Available.



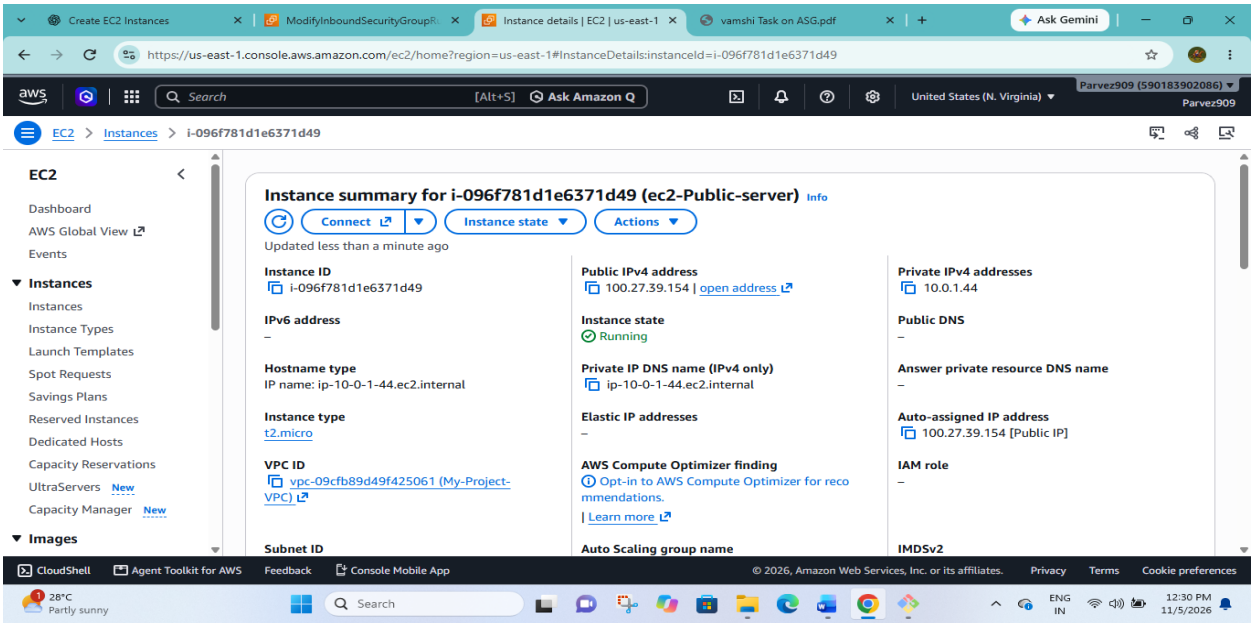
Next: Configure Private Route Table

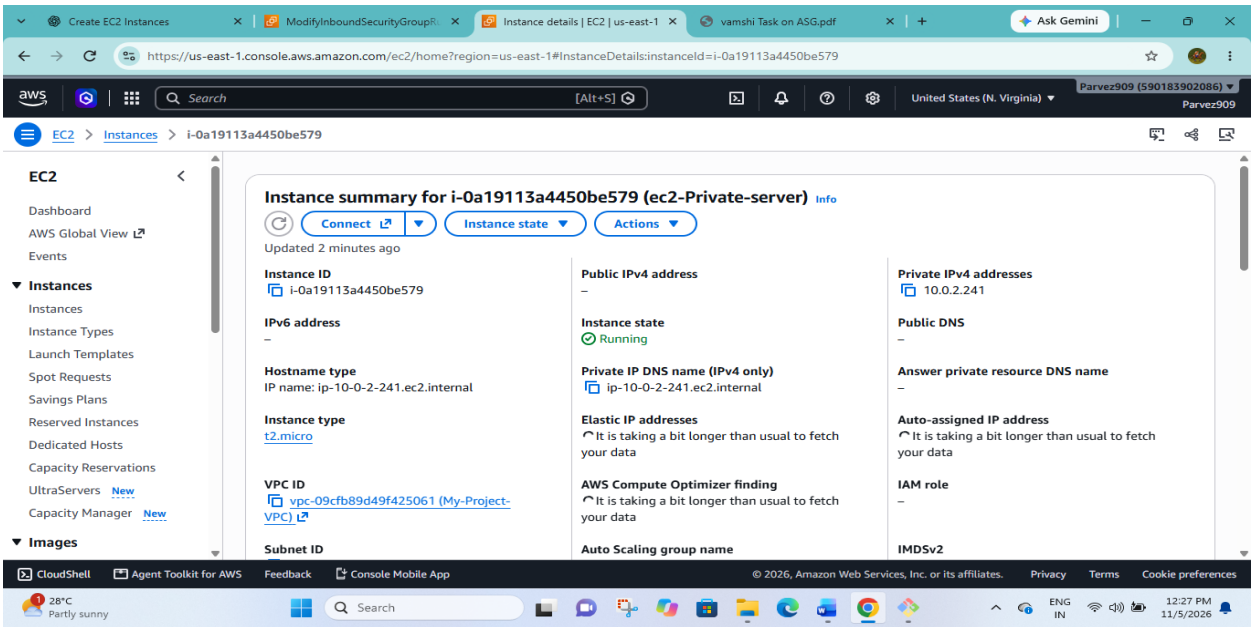
4. Go to:
5. VPC → Route Tables → Private-RT → Routes → Edit routes → Add route

Destination	Target
0.0.0.0/0	NAT Gateway → My-NAT-Gateway

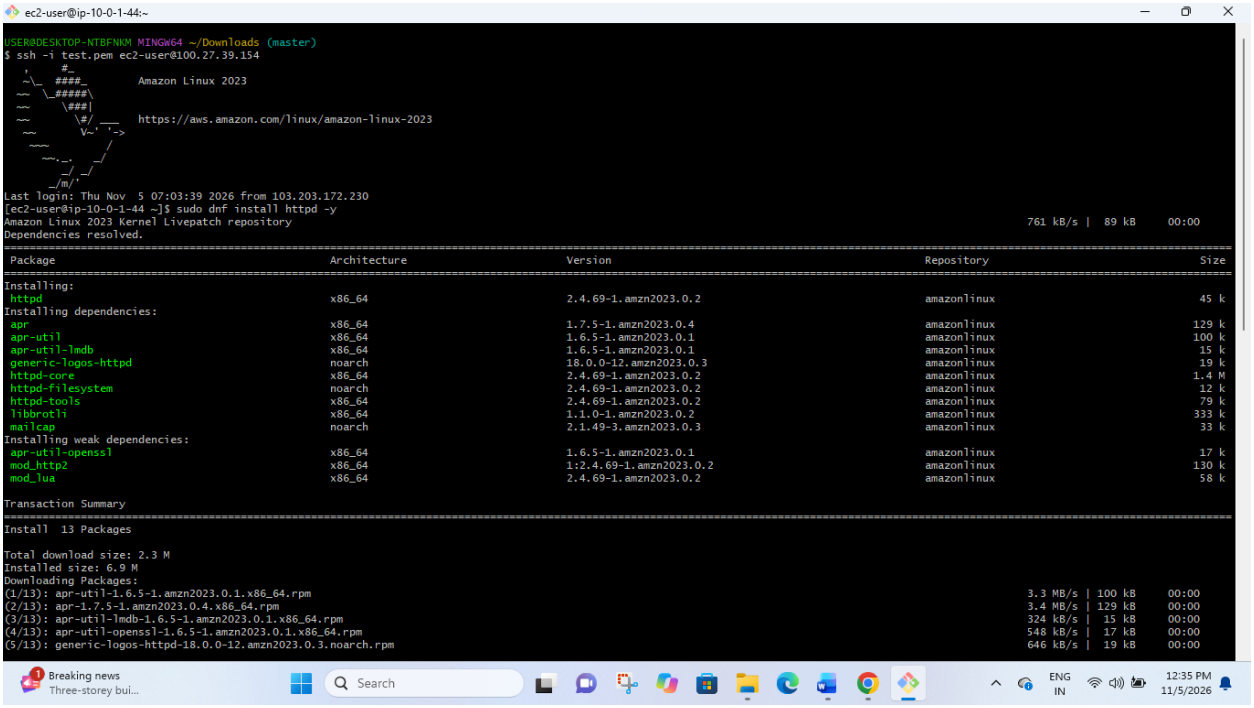


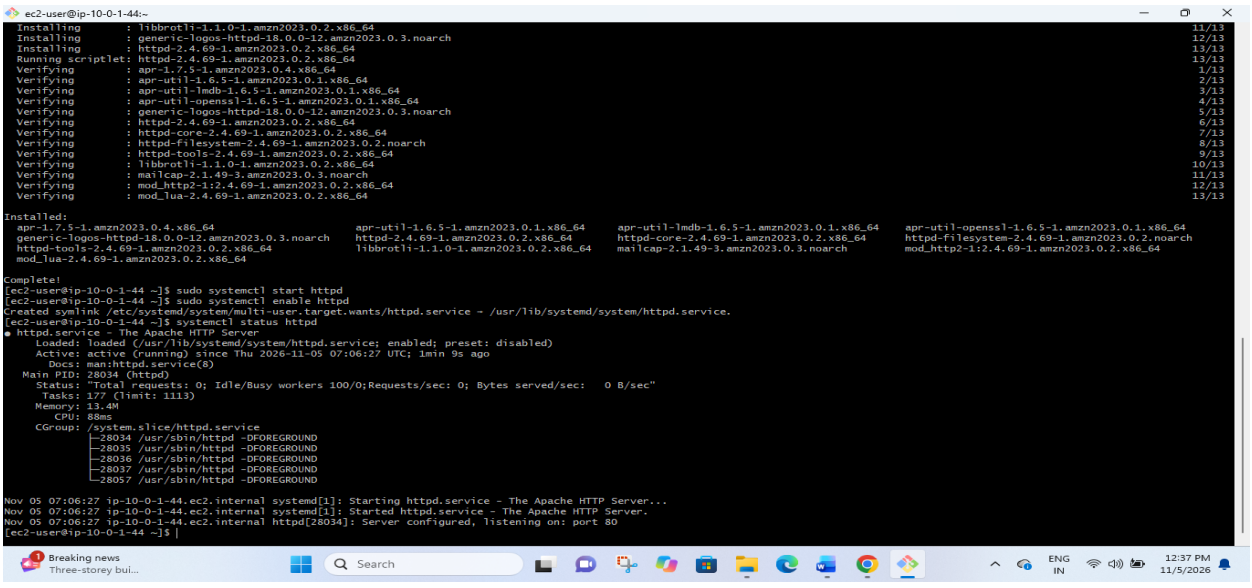
6. **Create two instances, one in the public subnet and one in the private subnet.**

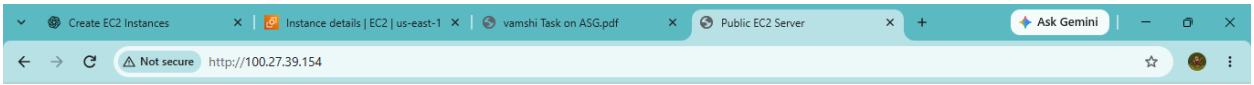




7. **Deploy Apache server on both EC2 instances with a sample index.html file.**





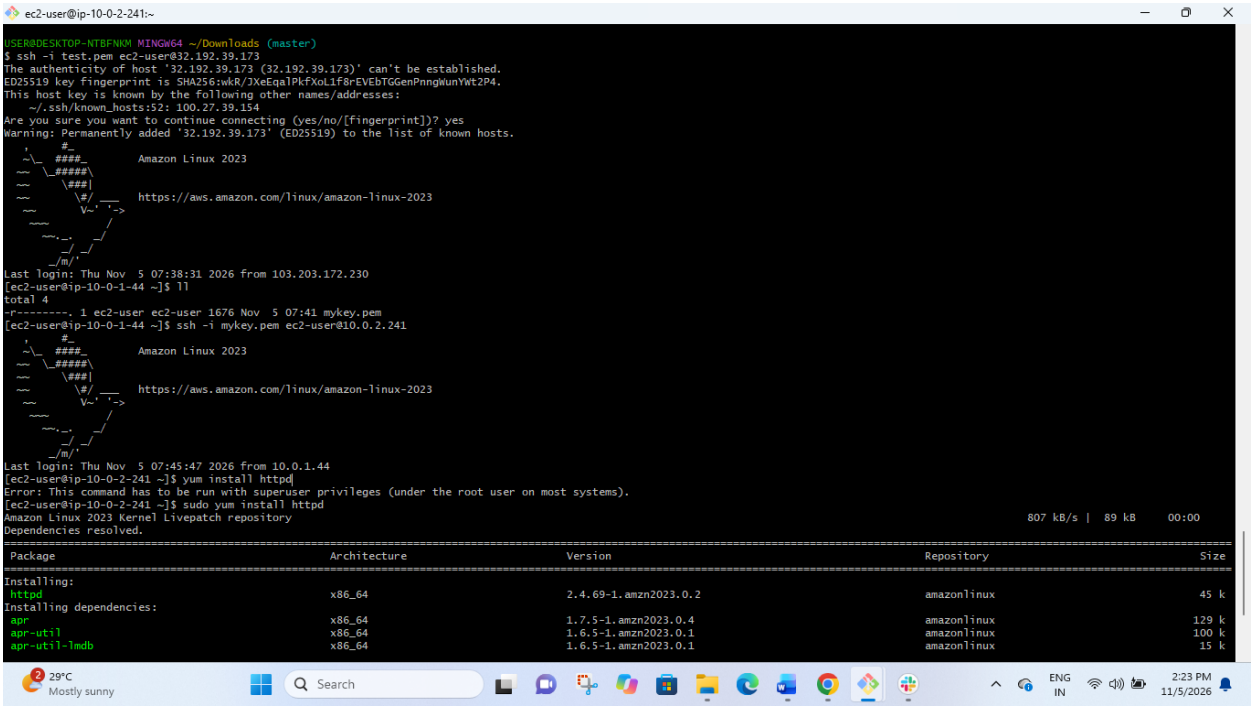


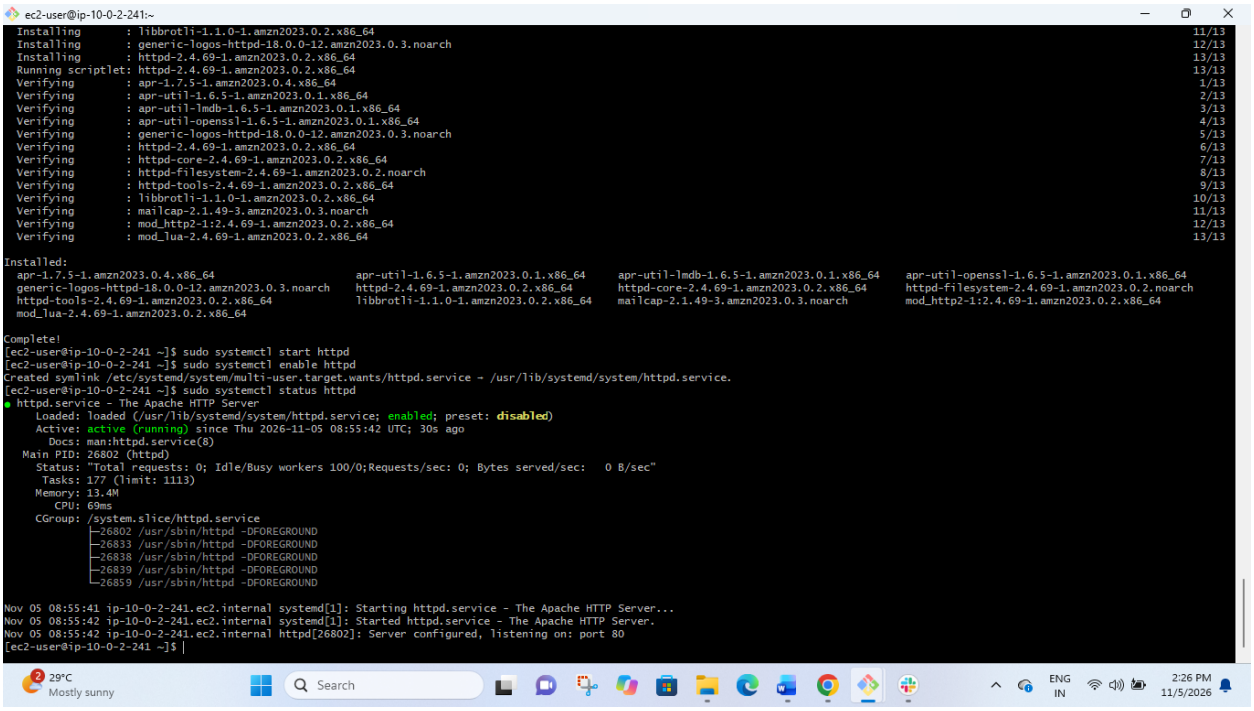
Hello Parvez!

This is Public EC2 Server is working



private ec2

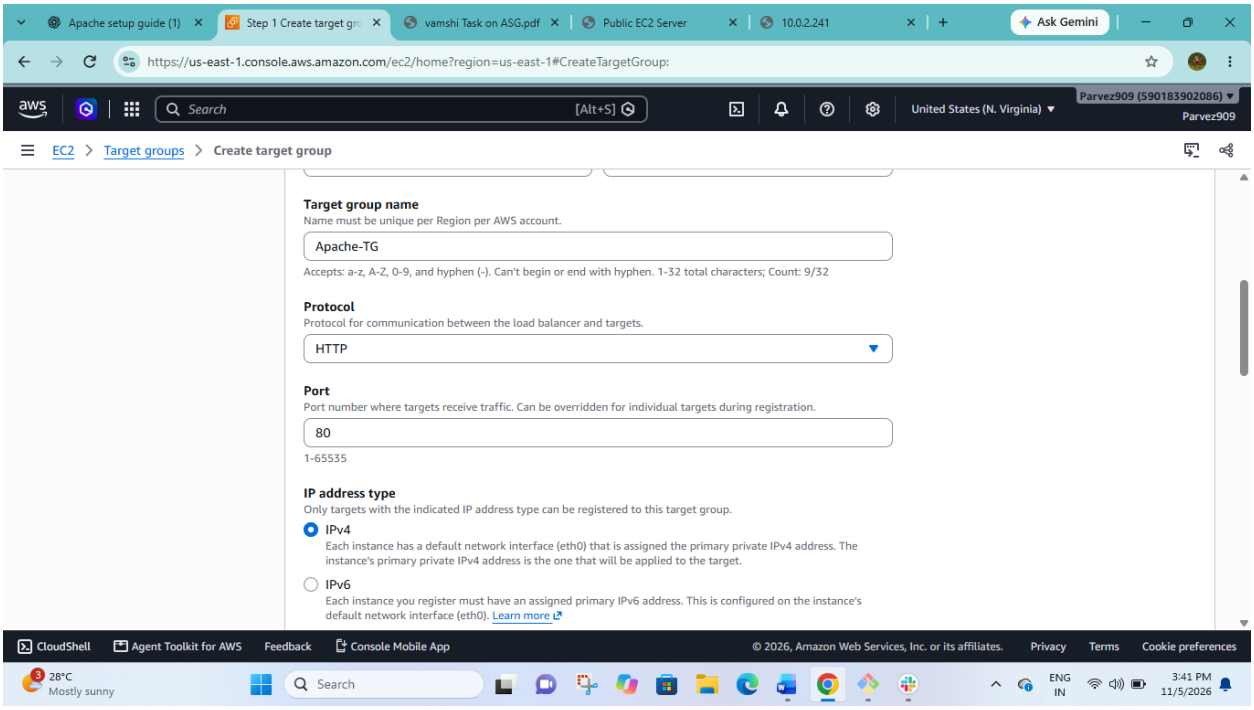


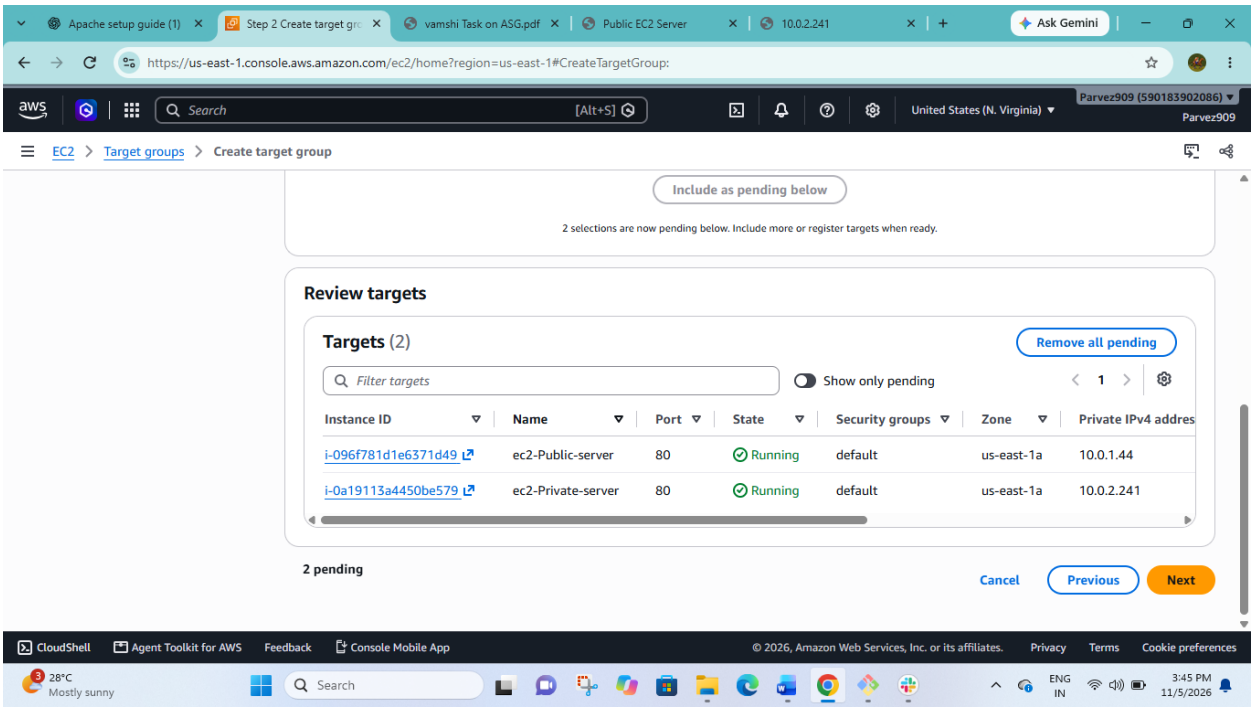


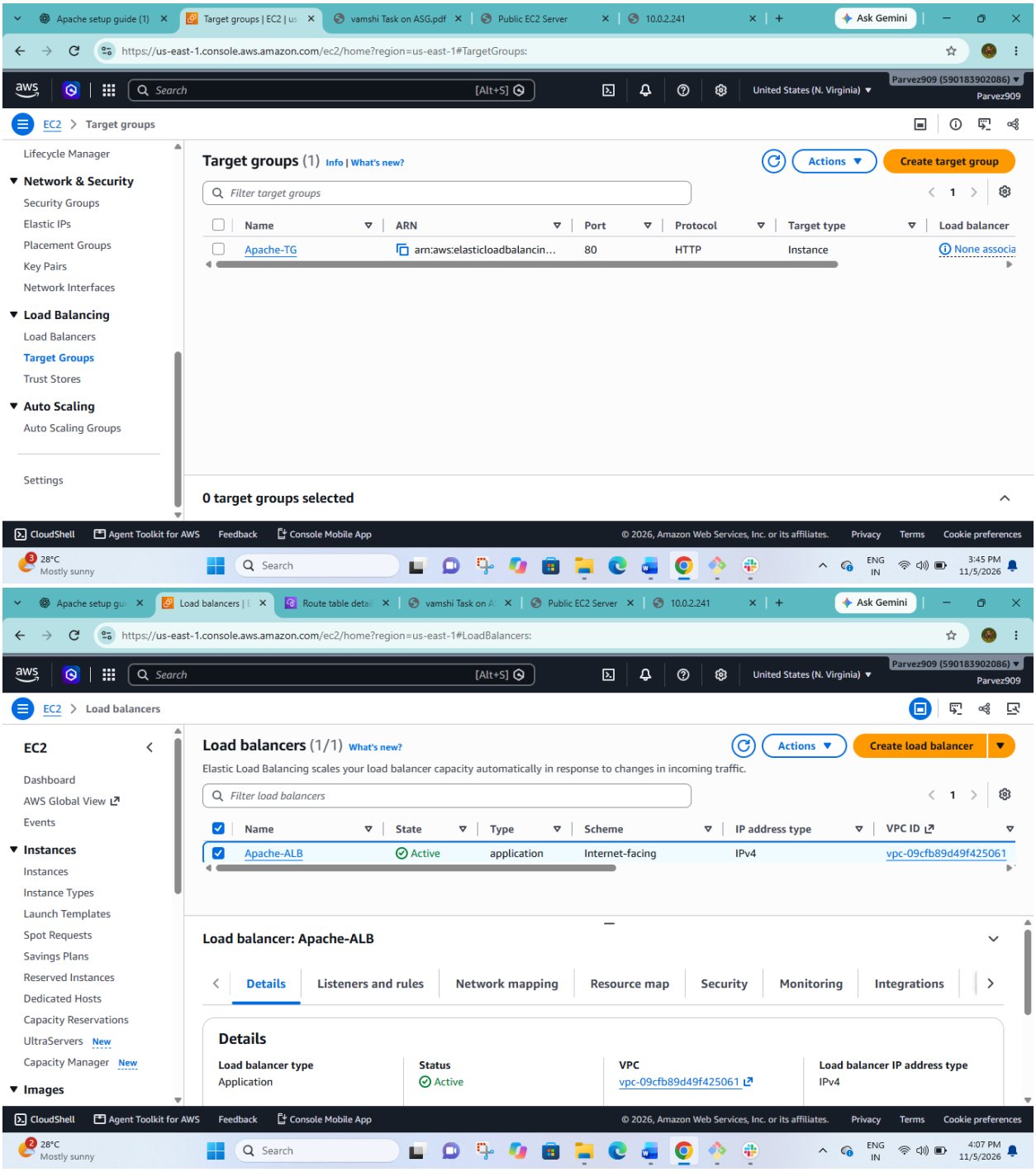

```
[ec2-user@ip-10-0-1-44 ~]$ ssh -i mykey.pem ec2-user@10.0.2.241
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Nov  5 08:48:45 2026 from 10.0.1.44
[ec2-user@ip-10-0-2-241 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-11-05 08:55:42 UTC; 18min ago
       Docs: man:httpd.service(8)
   Main PID: 26802 (httpd)
     Status: "Total requests: 3; Idle/Busy workers 100/0;Requests/sec: 0.00278; Bytes served/sec:   1 B/sec"
      Tasks: 177 (limit: 1113)
     Memory: 13.8M
        CPU: 679ms
     CGroup: /system.slice/httpd.service
             ├─26802 /usr/sbin/httpd -DFOREGROUND
             ├─26833 /usr/sbin/httpd -DFOREGROUND
             ├─26838 /usr/sbin/httpd -DFOREGROUND
             ├─26839 /usr/sbin/httpd -DFOREGROUND
             └─26859 /usr/sbin/httpd -DFOREGROUND

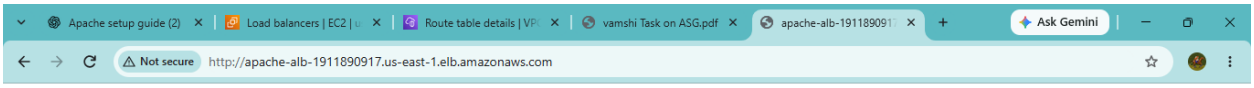
Nov 05 08:55:41 ip-10-0-2-241.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 05 08:55:42 ip-10-0-2-241.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 05 08:55:42 ip-10-0-2-241.ec2.internal httpd[26802]: Server configured, listening on: port 80
[ec2-user@ip-10-0-2-241 ~]$ netsat -n
-bash: netsat: command not found
[ec2-user@ip-10-0-2-241 ~]$ curl http://10.0.2.241:80
<h1>Hello Parvez!</h1>
<p>Apache running on Private EC2 perfectly working</p>
[ec2-user@ip-10-0-2-241 ~]$
```

8. **Create one application load balancer and attach it to both EC2 instances.**



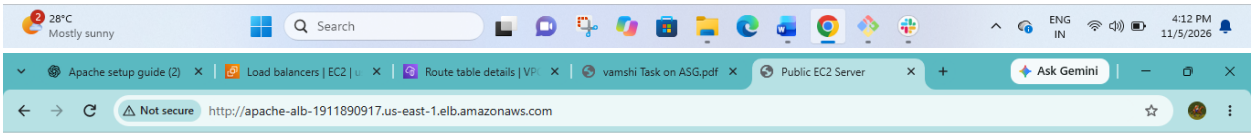



Apache setup guide (1)
Target groups | EC2 | us
vamshi Task on ASG.pdf
Public EC2 Server
10.0.2.241
Ask Gemini
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#TargetGroups:
Search
[Alt+S]
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
EC2 > Target groups
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Target groups (1) Info | What's new?
Actions
Create target group
Filter target groups
Name
ARN
Port
Protocol
Target type
Load balancer
Apache-TG
arn:aws:elasticloadbalancin...
80
HTTP
Instance
None associa
0 target groups selected
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
28°C
Mostly sunny
Search
ENG
IN
3:45 PM
11/5/2026
Load balancers | E
Route table detai
vamshi Task on A
Public EC2 Server
10.0.2.241
Ask Gemini
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancers:
Search
[Alt+S]
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
EC2 > Load balancers
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
Load balancers (1/1) What's new?
Actions
Create load balancer
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.
Filter load balancers
Name
State
Type
Scheme
IP address type
VPC ID
Apache-ALB
Active
application
Internet-facing
IPv4
vpc-09cfb89d49f425061
Load balancer: Apache-ALB
Details
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Details
Load balancer type
Application
Status
Active
VPC
vpc-09cfb89d49f425061
Load balancer IP address type
IPv4
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
28°C
Mostly sunny
Search
ENG
IN
4:07 PM
11/5/2026



Hello Parvez!

Apache running on Private EC2 perfectly working

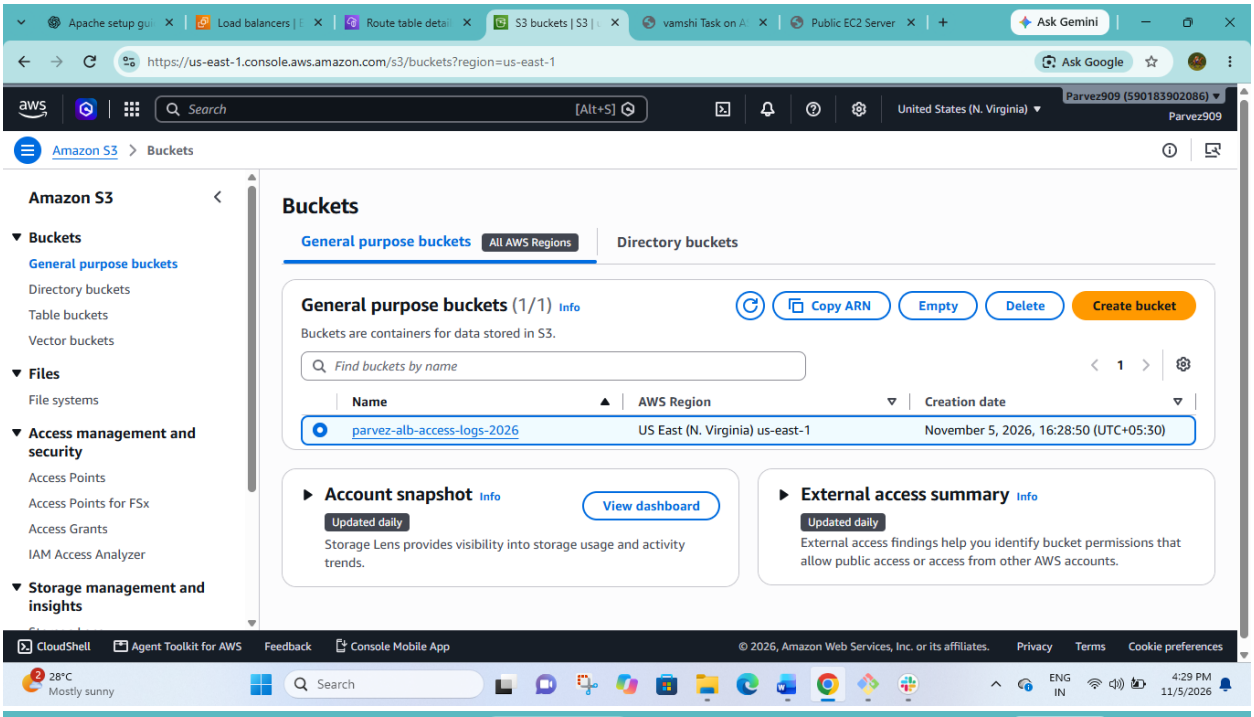


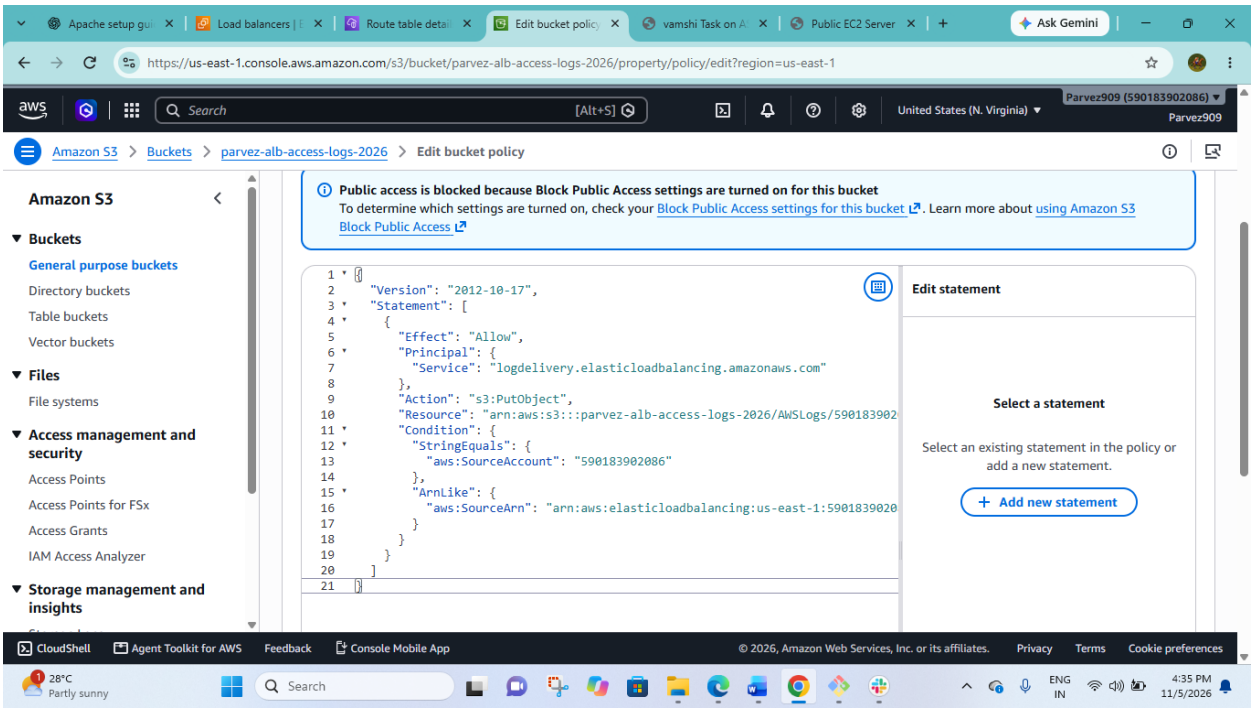
Hello Parvez!

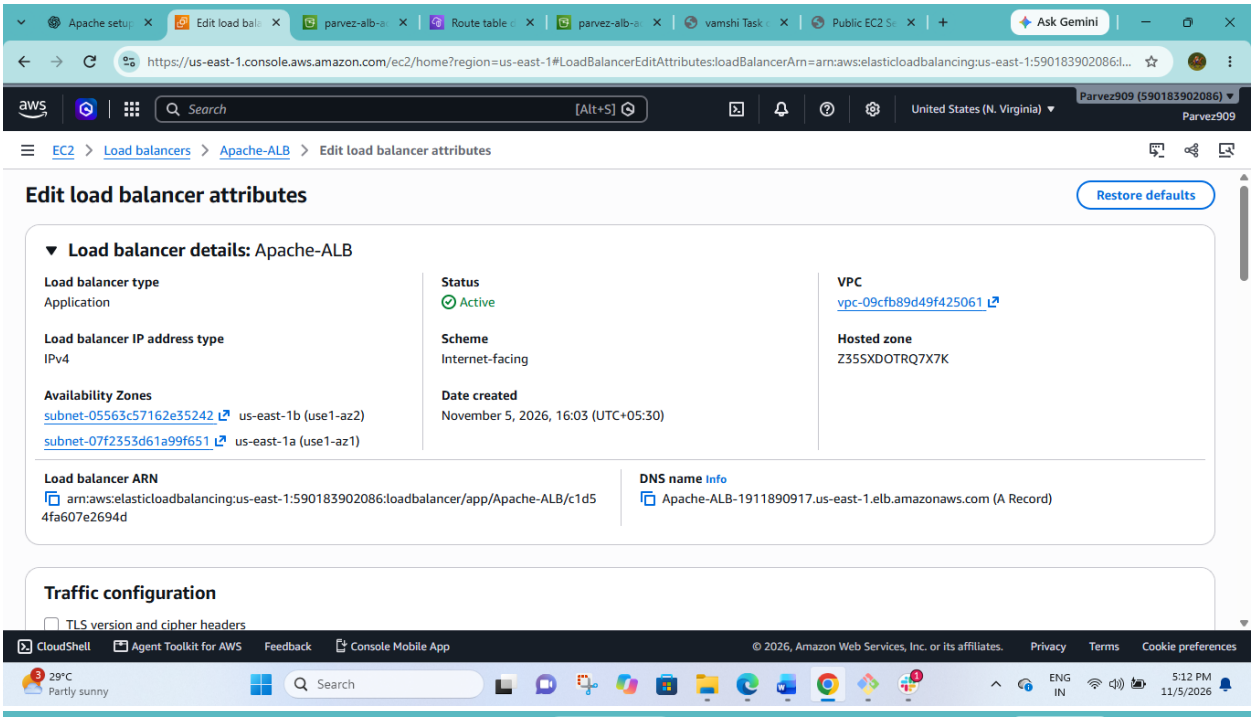
This is Public EC2 Server is ALB working

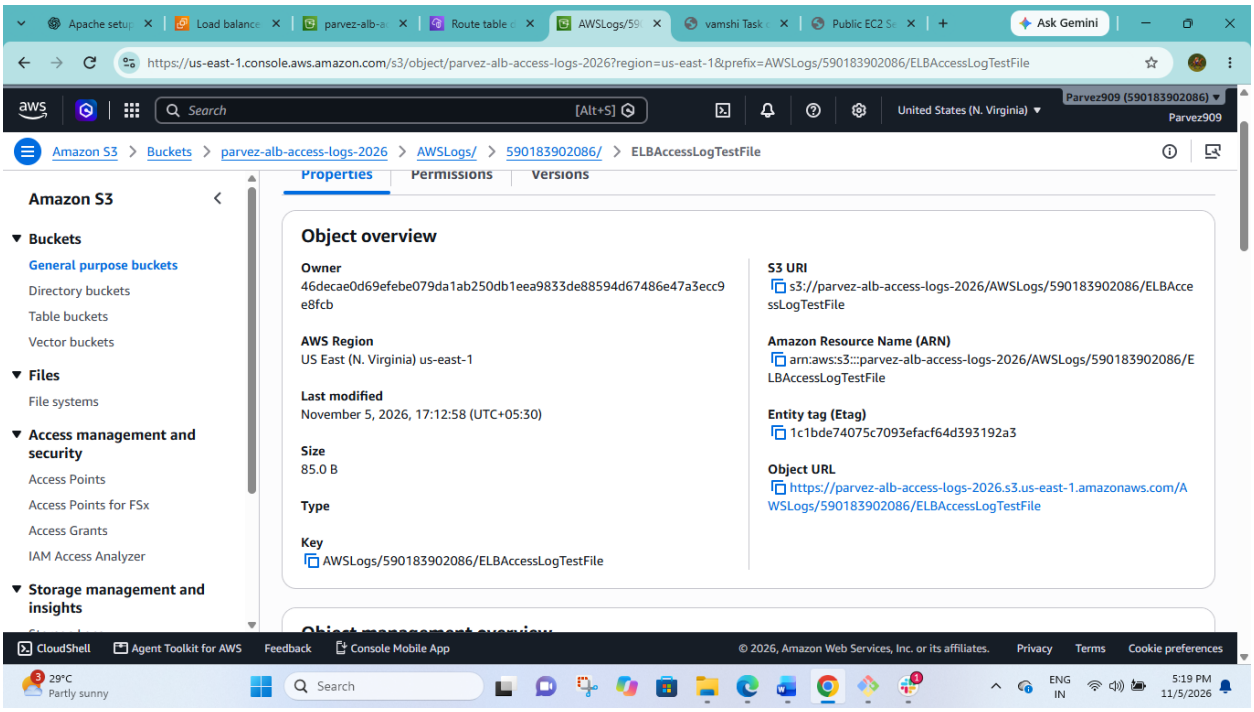


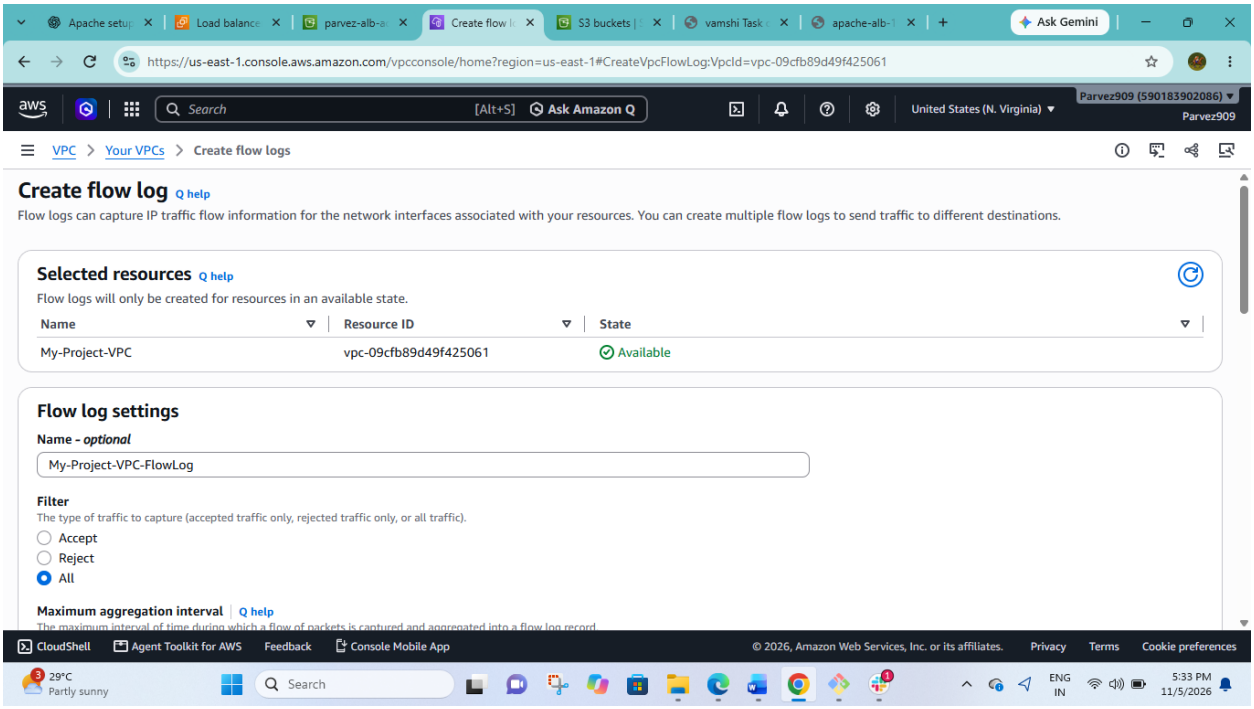
9. Store application load balancer logs in S3.



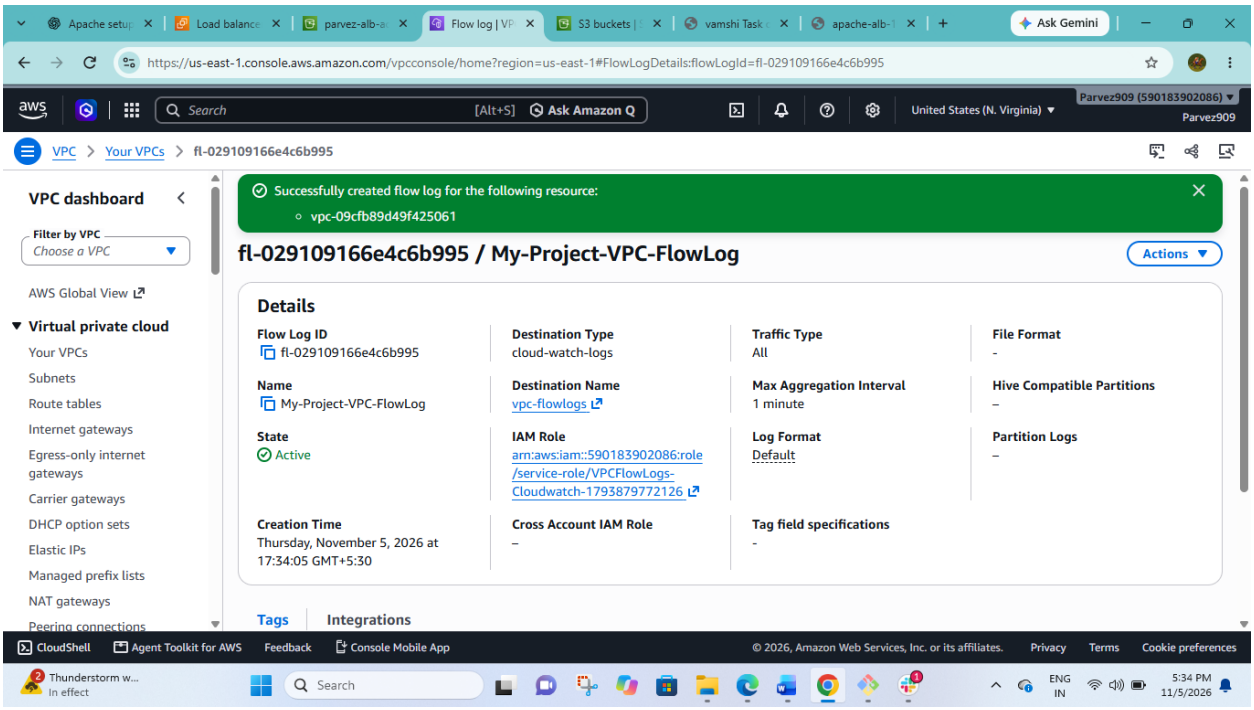


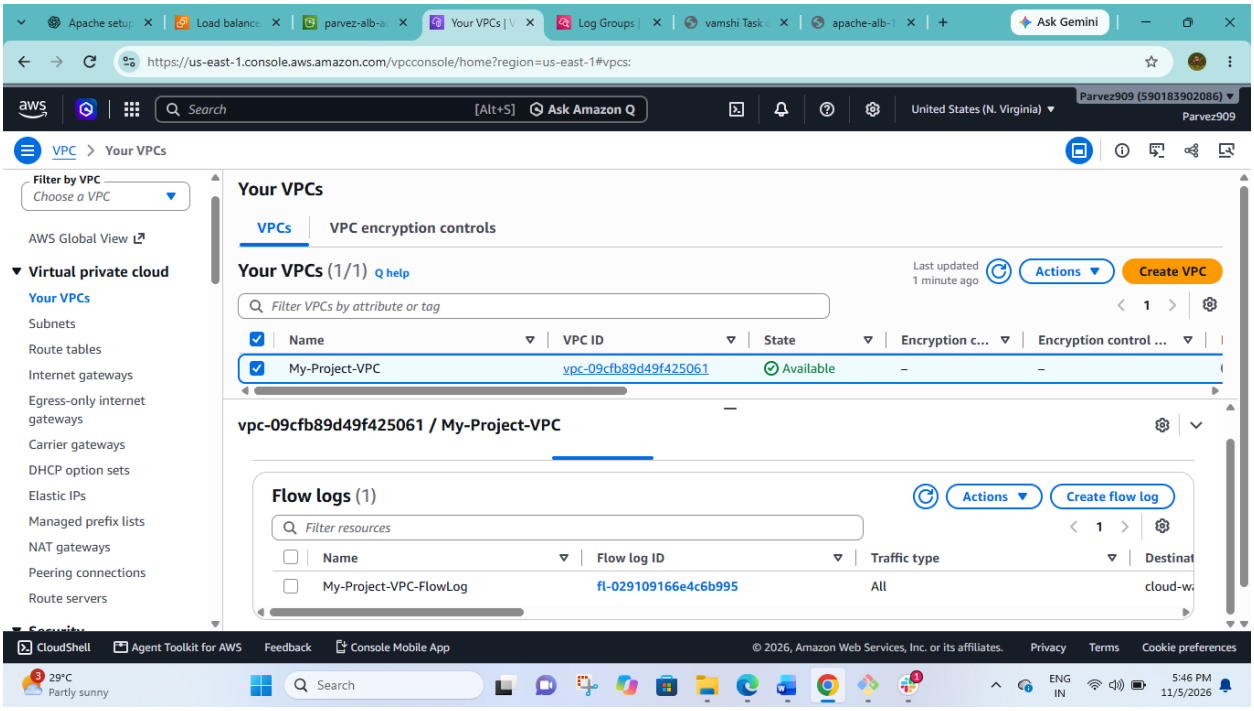

Edit load balancer attributes
Restore defaults
Load balancer details: Apache-ALB
Load balancer type
Application
Status
Active
VPC
vpc-09cfb89d49f425061
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-05563c57162e35242 us-east-1b (use1-az2)
subnet-07f2353d61a99f651 us-east-1a (use1-az1)
Date created
November 5, 2026, 16:03 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:590183902086:loadbalancer/app/Apache-ALB/c1d54fa607e2694d
DNS name Info
Apache-ALB-1911890917.us-east-1.elb.amazonaws.com (A Record)
Traffic configuration
TLS version and cipher headers

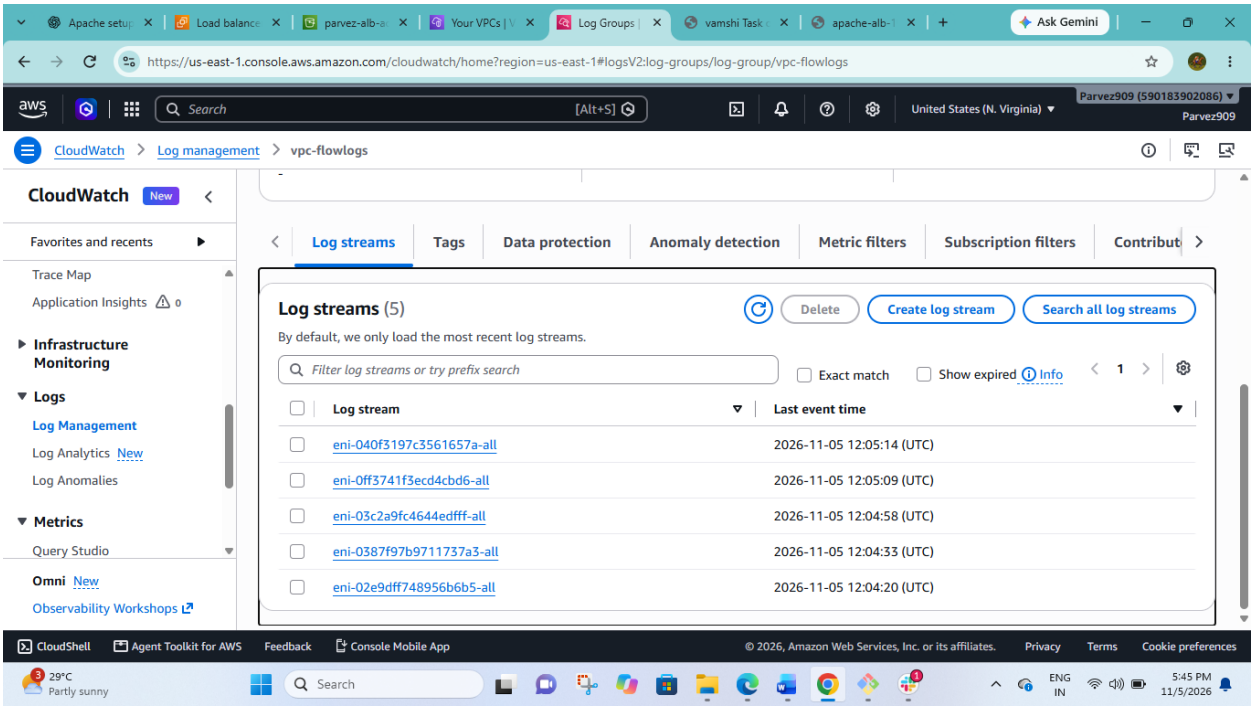

Amazon S3
Properties
Permissions
Versions
Object overview
Owner
46decae0d69efebe079da1ab250db1eea9833de88594d67486e47a3ecc9e8fcb
S3 URI
s3://parvez-alb-access-logs-2026/AWSLogs/590183902086/ELBAccessLogTestFile
AWS Region
US East (N. Virginia) us-east-1
Amazon Resource Name (ARN)
arn:aws:s3:::parvez-alb-access-logs-2026/AWSLogs/590183902086/ELBAccessLogTestFile
Last modified
November 5, 2026, 17:12:58 (UTC+05:30)
Entity tag (Etag)
1c1bde74075c7093efacf64d393192a3
Size
85.0 B
Object URL
https://parvez-alb-access-logs-2026.s3.us-east-1.amazonaws.com/AWSLogs/590183902086/ELBAccessLogTestFile
Type
Key
AWSLogs/590183902086/ELBAccessLogTestFile

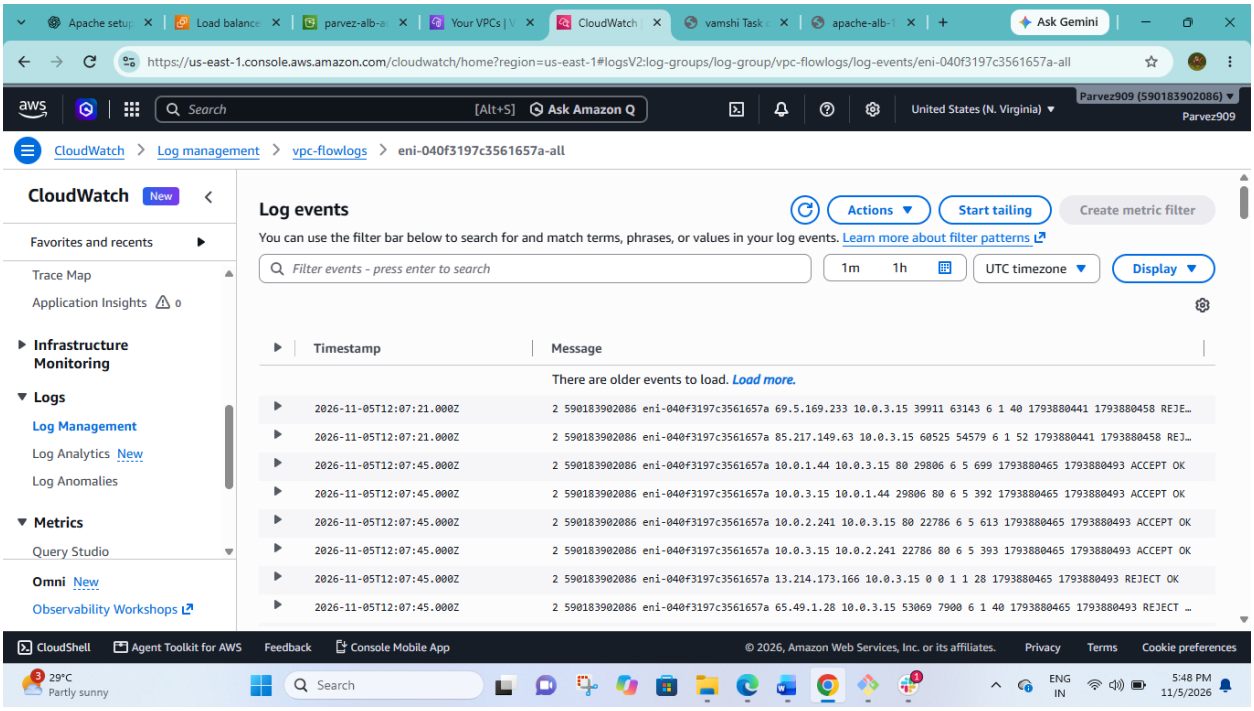


10. Store the VPC flow logs in a CloudWatch log group.

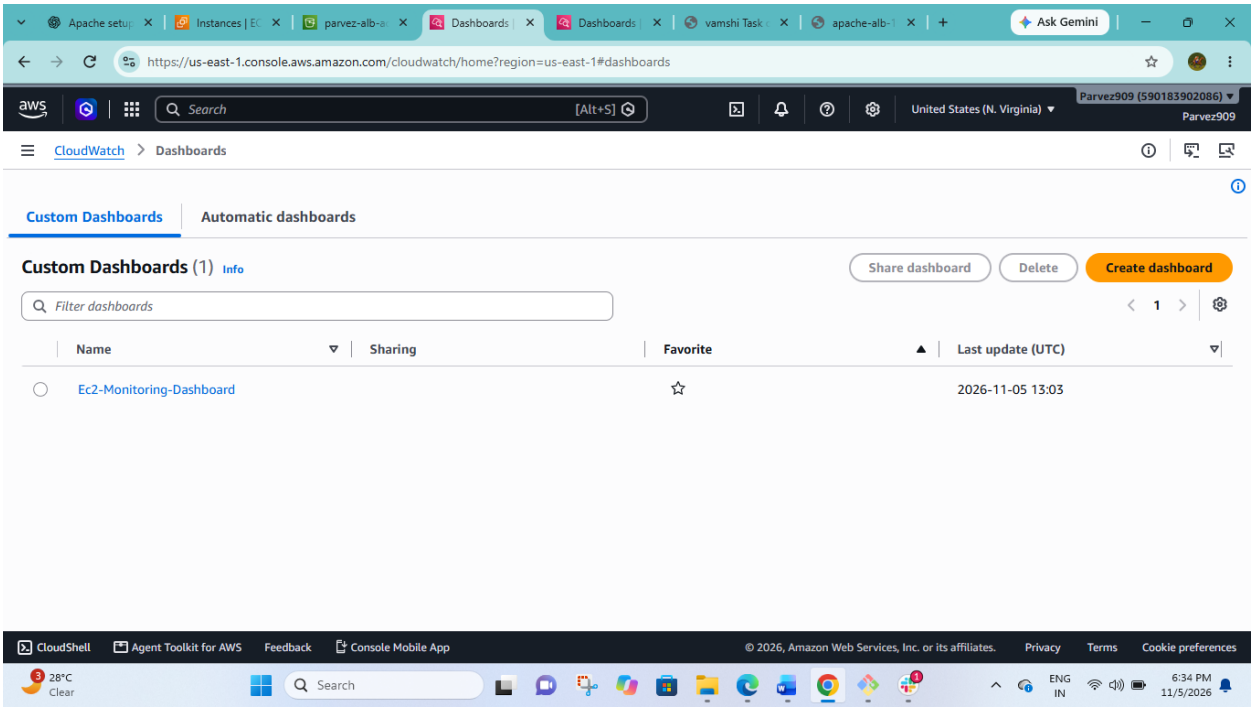


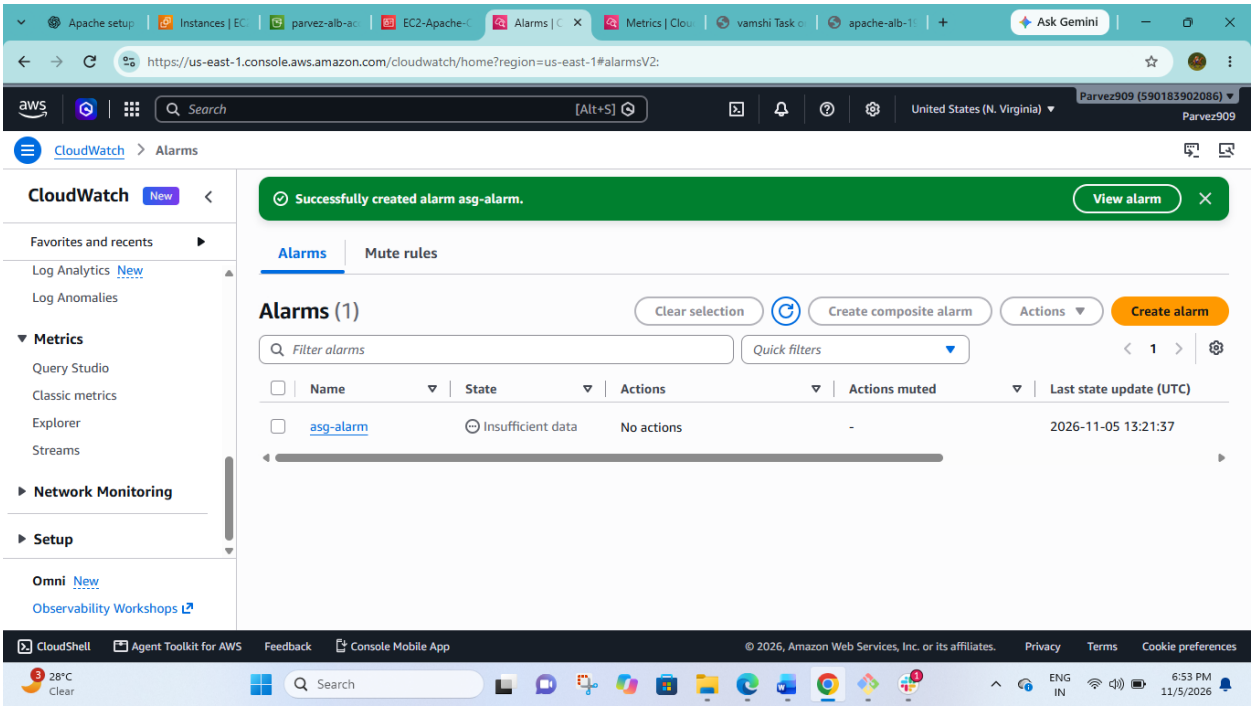

https://us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#vpcs:
VPC > Your VPCs
Your VPCs
VPCs
VPC encryption controls
Your VPCs (1/1)
Filter VPCs by attribute or tag
Name
VPC ID
State
My-Project-VPC
vpc-09cfb89d49f425061
Available
vpc-09cfb89d49f425061 / My-Project-VPC
Flow logs (1)
Create flow log
Filter resources
Flow log ID
Traffic type
My-Project-VPC-FlowLog
fl-029109166e4c6b995
All
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers


https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2:log-groups/log-group/vpc-flowlogs
CloudWatch > Log management > vpc-flowlogs
CloudWatch
Log streams
Tags
Data protection
Anomaly detection
Metric filters
Subscription filters
Log streams (5)
By default, we only load the most recent log streams.
Filter log streams or try prefix search
Log stream
Last event time
eni-040f3197c3561657a-all
2026-11-05 12:05:14 (UTC)
eni-0ff3741f3ecd4cbd6-all
2026-11-05 12:05:09 (UTC)
eni-03c2a9fc4644edfff-all
2026-11-05 12:04:58 (UTC)
eni-0387f97b9711737a3-all
2026-11-05 12:04:33 (UTC)
eni-02e9dff748956b6b5-all
2026-11-05 12:04:20 (UTC)
Log Management
Log Analytics
Log Anomalies
Metrics
Query Studio
Omni
Observability Workshops



11. Create monitoring dashboards to monitor CPU utilization and to monitor the Apache service.





12. If CPU utilization is more than 70%, then it should trigger auto scaling and launch new instance.

Create Launch Template

Launch Template: Apache-Launch-Template ✓

AMI: Amazon Linux 2023 ✓

Instance: t2.micro ✓

Key Pair: test ✓

Subnet: Don't include ✓

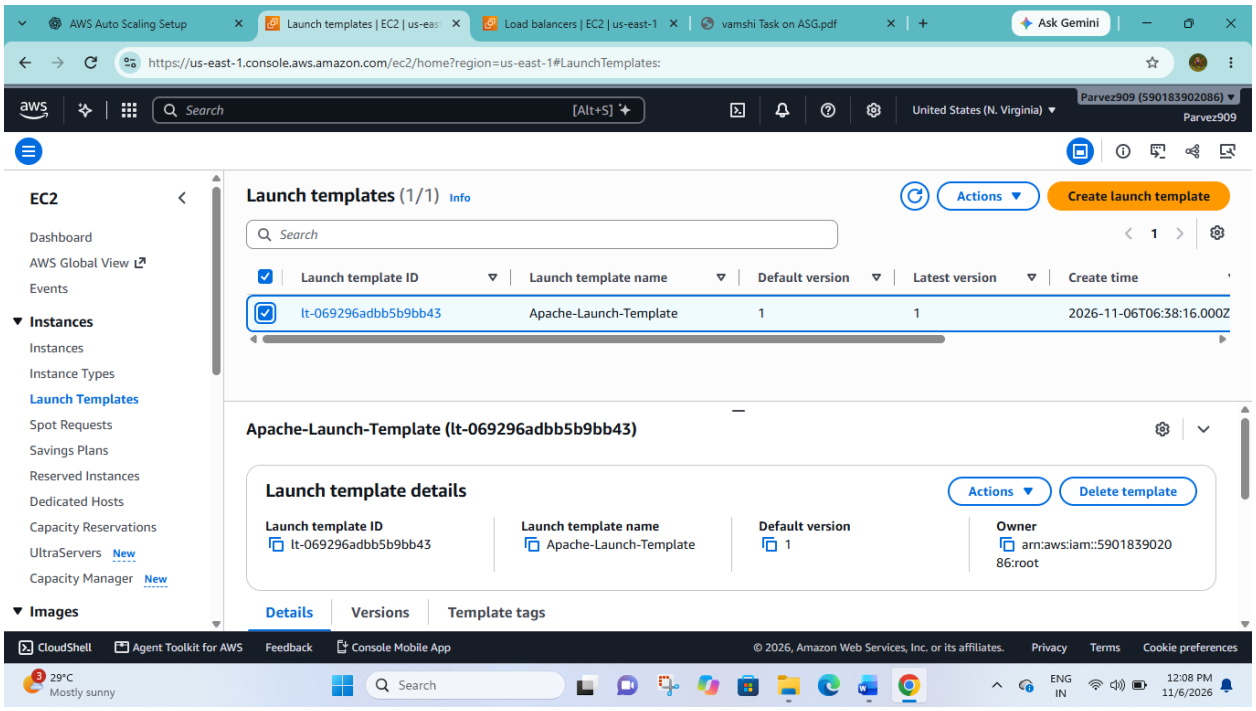
Security Group: Apache EC2 Security Group ✓

HTTP Port: 80 ✓

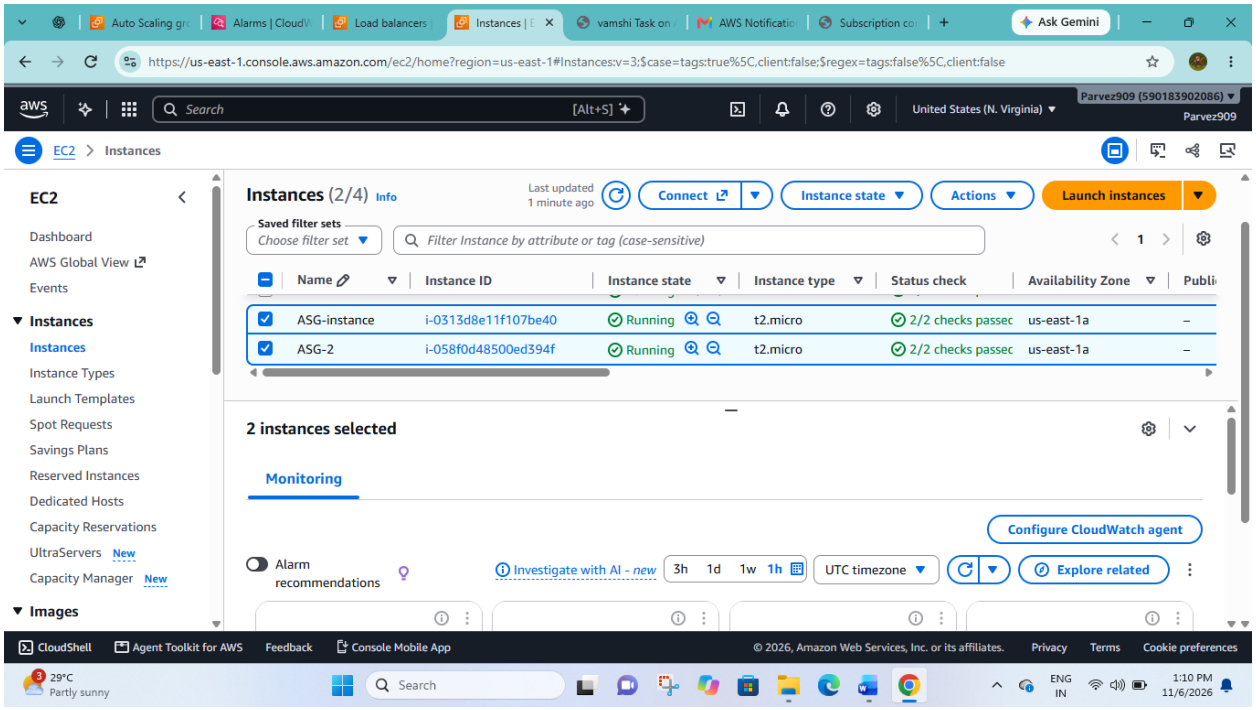
Source: ALB-SG ✓

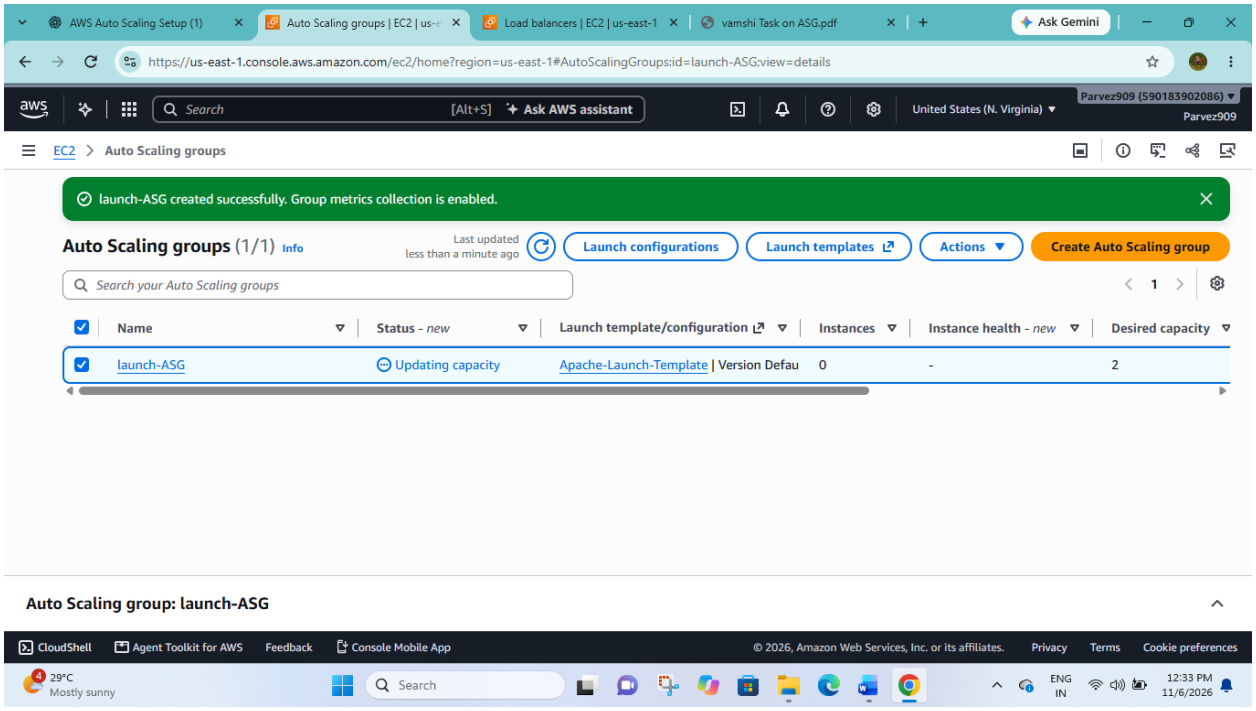
User Data: Apache installation script ✓

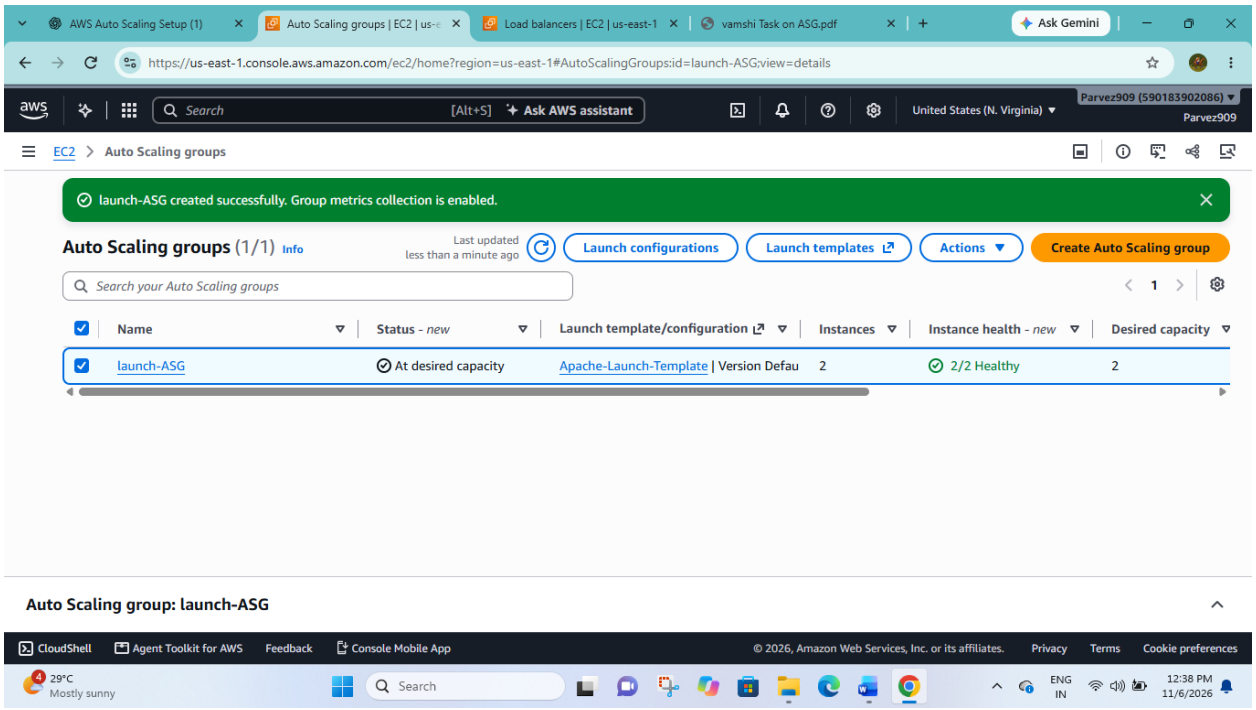
Launch Template Created Successfully



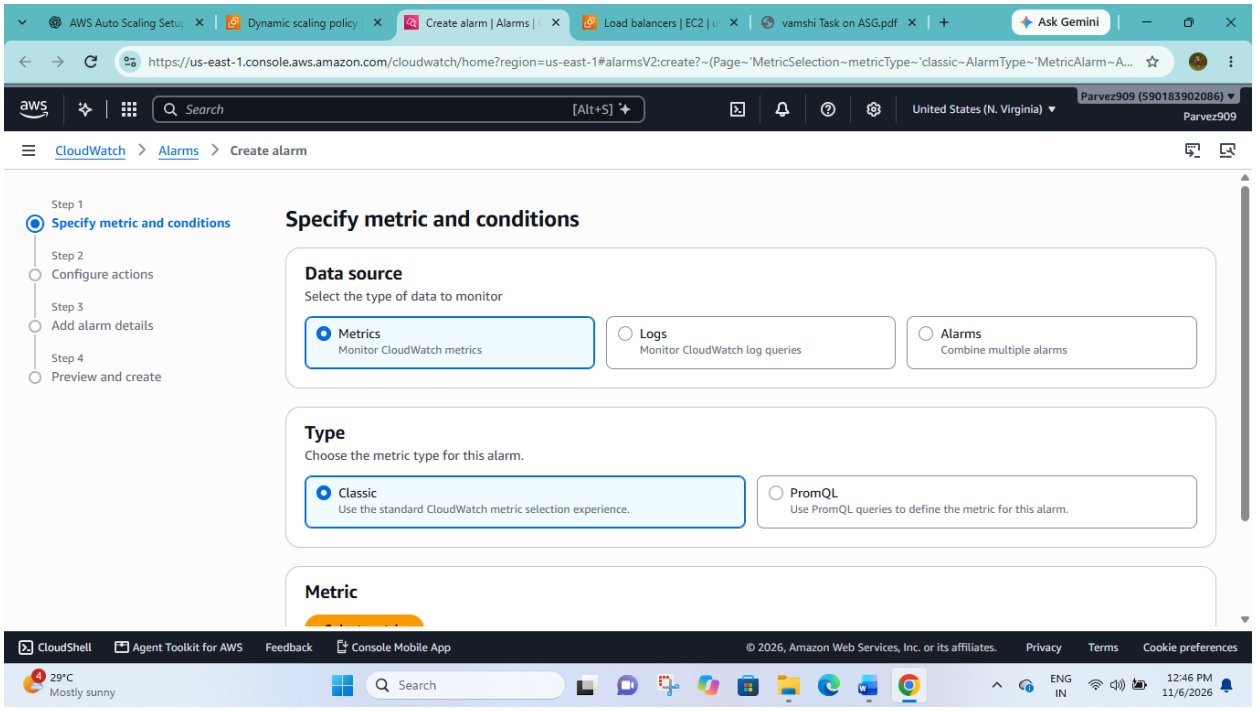
Create Auto Scaling Group (Minimum 1, Desired 2, Maximum 5).

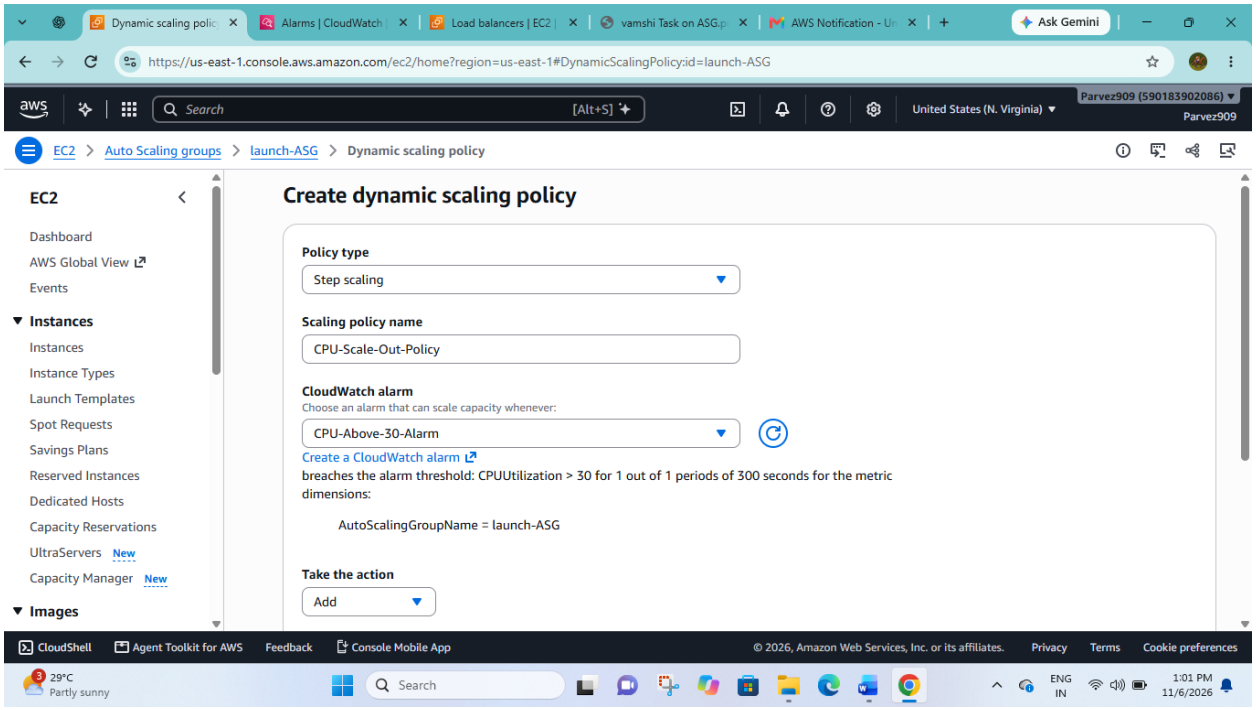


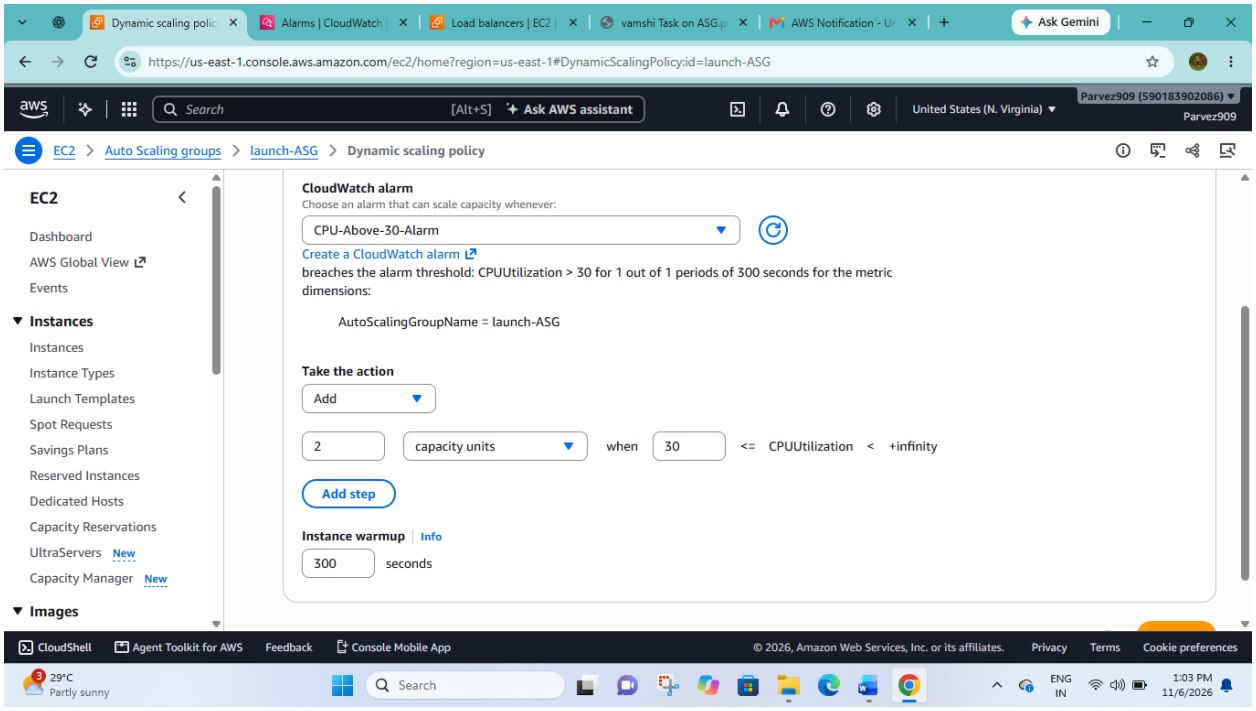


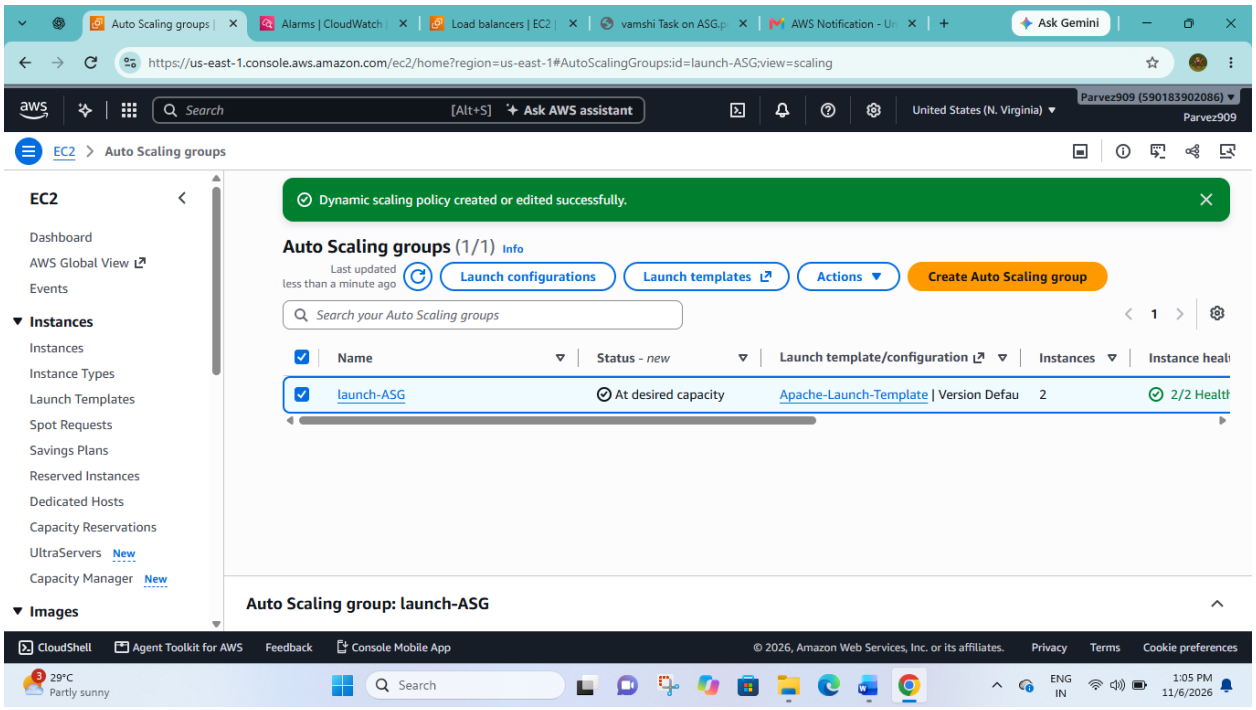


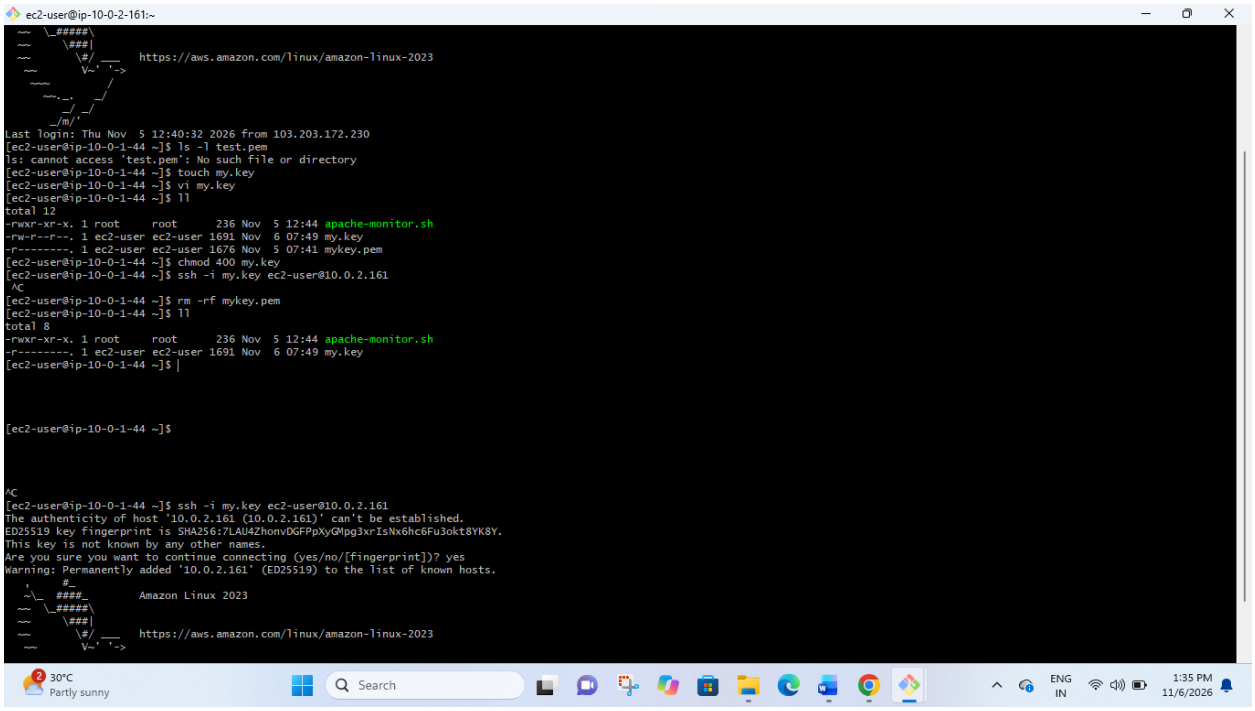
Create a CloudWatch alarm

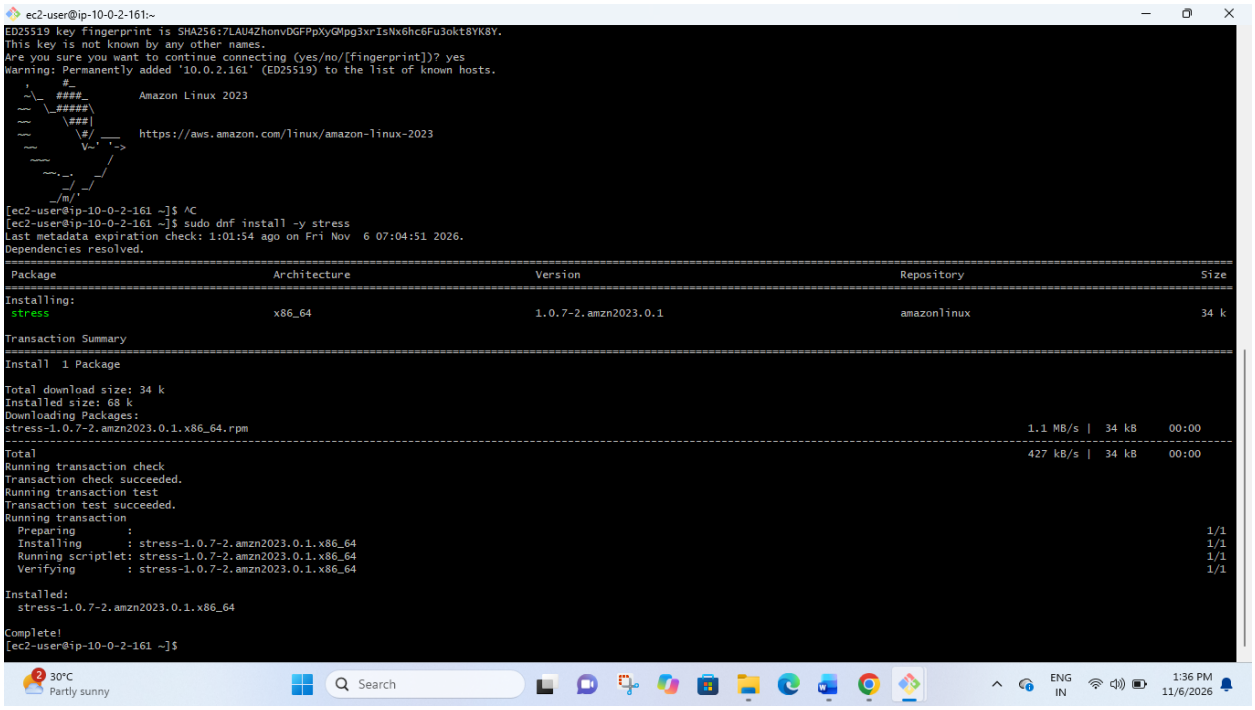

Create alarm | Alarms |
https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:create?~(Page~'MetricSelection~metricType~'classic~AlarmType~'MetricAlarm~A...
Search
[Alt+S]
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
CloudWatch > Alarms > Create alarm
Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Specify metric and conditions
Data source
Select the type of data to monitor
Metrics
Monitor CloudWatch metrics
Logs
Monitor CloudWatch log queries
Alarms
Combine multiple alarms
Type
Choose the metric type for this alarm.
Classic
Use the standard CloudWatch metric selection experience.
PromQL
Use PromQL queries to define the metric for this alarm.
Metric
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Feedback
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29°C
Mostly sunny
Search
ENG
IN
12:46 PM
11/6/2026

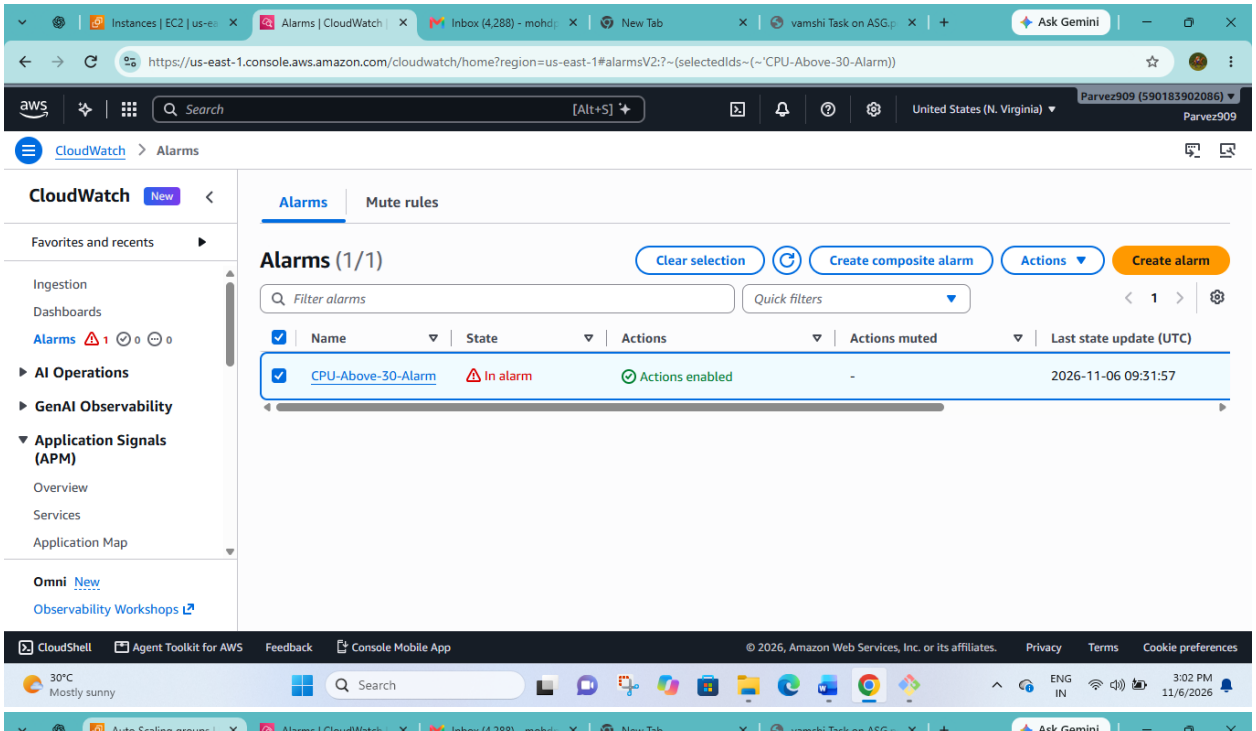

Dynamic scaling polic
Alarms | CloudWatch
Load balancers | EC2
vamshi Task on ASG.p
AWS Notification - Ur
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#DynamicScalingPolicy:id=launch-ASG
Search
[Alt+S]
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
EC2 > Auto Scaling groups > launch-ASG > Dynamic scaling policy
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
Create dynamic scaling policy
Policy type
Step scaling
Scaling policy name
CPU-Scale-Out-Policy
CloudWatch alarm
Choose an alarm that can scale capacity whenever:
CPU-Above-30-Alarm
Create a CloudWatch alarm
breaches the alarm threshold: CPUUtilization > 30 for 1 out of 1 periods of 300 seconds for the metric dimensions:
AutoScalingGroupName = launch-ASG
Take the action
Add
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Search
ENG
IN
1:01 PM
11/6/2026

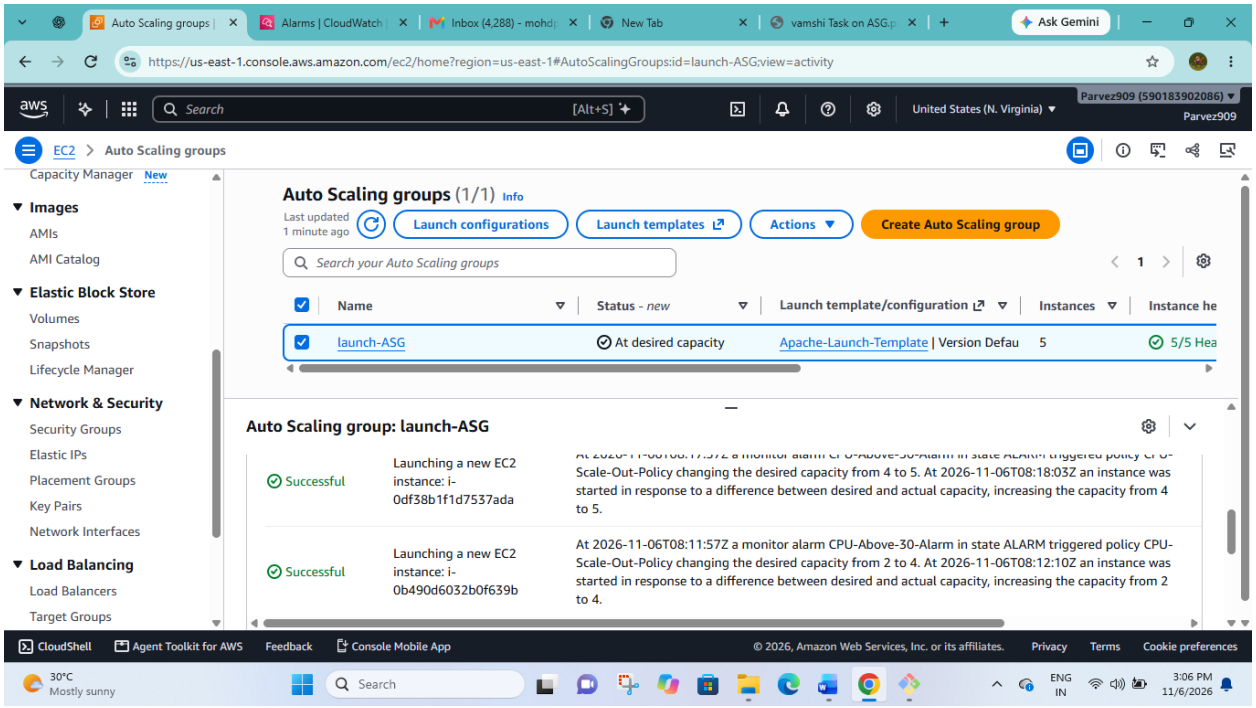

Dynamic scaling polic
Alarms | CloudWatch
Load balancers | EC2
vamshi Task on ASG.p
AWS Notification - Ur
Ask Gemini
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#DynamicScalingPolicy:id=launch-ASG
Search
[Alt+S]
Ask AWS assistant
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
EC2 > Auto Scaling groups > launch-ASG > Dynamic scaling policy
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
CloudWatch alarm
Choose an alarm that can scale capacity whenever:
CPU-Above-30-Alarm
Create a CloudWatch alarm
breaches the alarm threshold: CPUUtilization > 30 for 1 out of 1 periods of 300 seconds for the metric
dimensions:
AutoScalingGroupName = launch-ASG
Take the action
Add
2
capacity units
when
30
<= CPUUtilization < +infinity
Add step
Instance warmup | Info
300
seconds
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Search
ENG
IN
1:03 PM
11/6/2026


Auto Scaling groups
Alarms | CloudWatch
Load balancers | EC2
vamshi Task on ASG.p
AWS Notification - Ur
Ask Gemini
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#AutoScalingGroups:id=launch-ASG;view=scaling
Search
[Alt+S]
Ask AWS assistant
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
EC2 > Auto Scaling groups
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
Dynamic scaling policy created or edited successfully.
Auto Scaling groups (1/1) Info
Last updated less than a minute ago
Launch configurations
Launch templates
Actions
Create Auto Scaling group
Search your Auto Scaling groups
1
Name
Status - new
Launch template/configuration
Instances
Instance heal
launch-ASG
At desired capacity
Apache-Launch-Template | Version Defau
2
2/2 Healt
Auto Scaling group: launch-ASG
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Search
ENG
IN
1:05 PM
11/6/2026


ec2-user@ip-10-0-2-161:~
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Thu Nov 5 12:40:32 2026 from 103.203.172.230
[ec2-user@ip-10-0-1-44 ~]$ ls -l test.pem
ls: cannot access 'test.pem': No such file or directory
[ec2-user@ip-10-0-1-44 ~]$ touch my.key
[ec2-user@ip-10-0-1-44 ~]$ vi my.key
[ec2-user@ip-10-0-1-44 ~]$ ll
total 12
-rwxr-xr-x. 1 root root 236 Nov 5 12:44 apache-monitor.sh
-rw-r--r--. 1 ec2-user ec2-user 1691 Nov 6 07:49 my.key
-r--------. 1 ec2-user ec2-user 1676 Nov 5 07:41 mykey.pem
[ec2-user@ip-10-0-1-44 ~]$ chmod 400 my.key
[ec2-user@ip-10-0-1-44 ~]$ ssh -i my.key ec2-user@10.0.2.161
^C
[ec2-user@ip-10-0-1-44 ~]$ rm -rf mykey.pem
[ec2-user@ip-10-0-1-44 ~]$ ll
total 8
-rwxr-xr-x. 1 root root 236 Nov 5 12:44 apache-monitor.sh
-r--------. 1 ec2-user ec2-user 1691 Nov 6 07:49 my.key
[ec2-user@ip-10-0-1-44 ~]$
[ec2-user@ip-10-0-1-44 ~]$
^C
[ec2-user@ip-10-0-1-44 ~]$ ssh -i my.key ec2-user@10.0.2.161
The authenticity of host '10.0.2.161 (10.0.2.161)' can't be established.
ED25519 key fingerprint is SHA256:7LAU4ZhonvDGFPpXyGMpg3xrIsNx6hc6Fu3okt8YK8Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.161' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
30°C
Partly sunny
Search
ENG
IN
1:35 PM
11/6/2026


ec2-user@ip-10-0-2-161:~
ED25519 key fingerprint is SHA256:7LAU4ZhonvDGFPpXyGMpg3xrIsNx6hc6Fu3okt8YK8Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.161' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-0-2-161 ~]$ ^C
[ec2-user@ip-10-0-2-161 ~]$ sudo dnf install -y stress
Last metadata expiration check: 1:01:54 ago on Fri Nov 6 07:04:51 2026.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
stress x86_64 1.0.7-2.amzn2023.0.1 amazonlinux 34 k
Transaction Summary
Install 1 Package
Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kB 00:00
Total 427 kB/s | 34 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Installing : stress-1.0.7-2.amzn2023.0.1.x86_64
Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64
Verifying : stress-1.0.7-2.amzn2023.0.1.x86_64
Installed:
stress-1.0.7-2.amzn2023.0.1.x86_64
Complete!
[ec2-user@ip-10-0-2-161 ~]$
30°C
Partly sunny
Search
ENG
IN
1:36 PM
11/6/2026


Alarms | CloudWatch
https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:?~(selectedIds~('CPU-Above-30-Alarm))
Search
United States (N. Virginia)
Parvez909
CloudWatch > Alarms
CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Overview
Services
Application Map
Omni New
Observability Workshops
Alarms
Mute rules
Alarms (1/1)
Clear selection
Create composite alarm
Actions
Create alarm
Filter alarms
Quick filters
Name
State
Actions
Actions muted
Last state update (UTC)
CPU-Above-30-Alarm
In alarm
Actions enabled
2026-11-06 09:31:57
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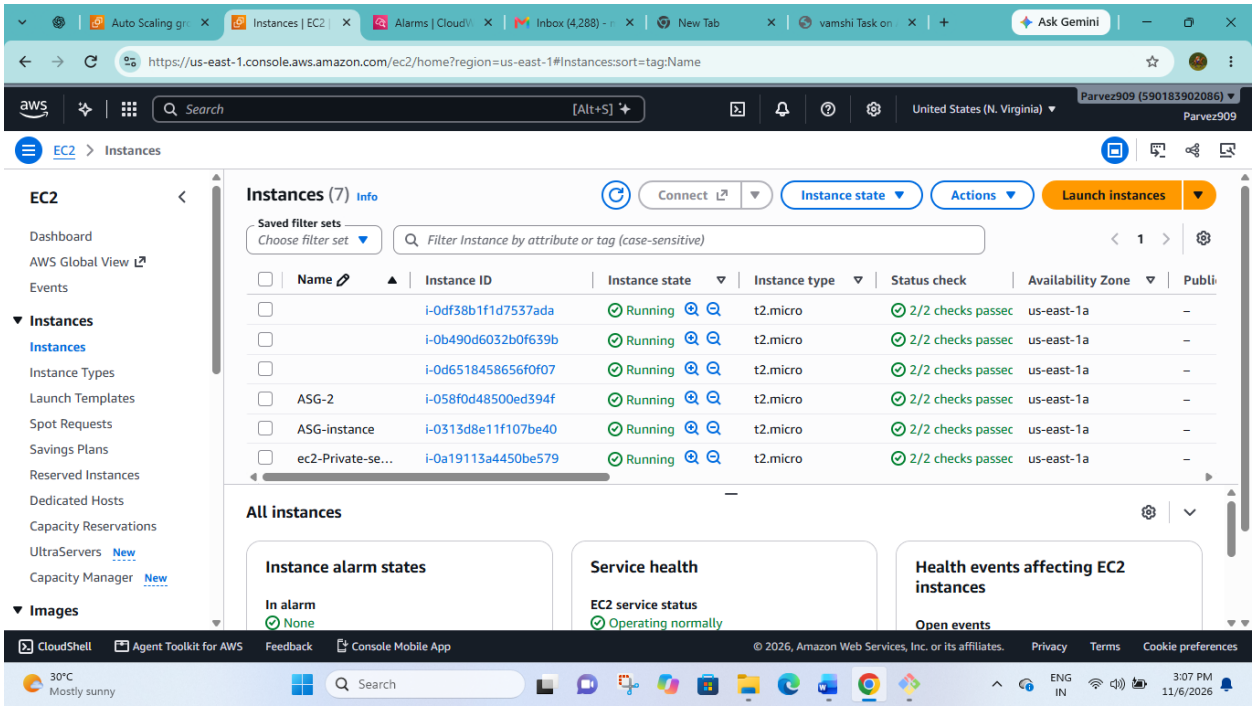

Auto Scaling groups
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#AutoScalingGroups:id=launch-ASG;view=activity
EC2 > Auto Scaling groups
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Auto Scaling groups (1/1) Info
Last updated 1 minute ago
Launch configurations
Launch templates
Actions
Create Auto Scaling group
Search your Auto Scaling groups
Name
Status - new
Launch template/configuration
Instances
launch-ASG
At desired capacity
Apache-Launch-Template | Version Defau
5
5/5 Hea
Auto Scaling group: launch-ASG
Successful
Launching a new EC2 instance: i-0df38b1f1d7537ada
Scale-Out-Policy changing the desired capacity from 4 to 5. At 2026-11-06T08:18:03Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 4 to 5.
Successful
Launching a new EC2 instance: i-0b490d6032b0f639b
At 2026-11-06T08:11:57Z a monitor alarm CPU-Above-30-Alarm in state ALARM triggered policy CPU-Scale-Out-Policy changing the desired capacity from 2 to 4. At 2026-11-06T08:12:10Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 2 to 4.
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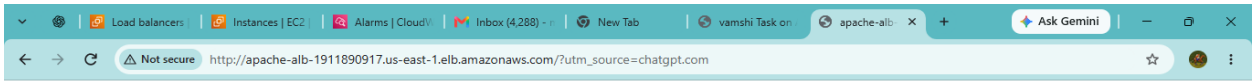
AWS Auto Scaling Assignment – Progress

Mostly completed

Step	Implementation	Status
1	Create EC2 Launch Template	Complete
2	Select Amazon Linux 2023	Complete
3	Select instance type (t2.micro)	Complete
4	Configure Apache security group	Complete
5	Add User Data to install/start Apache	Complete
6	Create Auto Scaling Group	Complete
7	Select Launch Template	Complete
8	Select VPC	Complete
9	Select private subnets in multiple AZs	Pending
10–13	Min 1, Desired 2, Max 5	Complete
14	Attach Apache target group	Complete
15	Enable ELB health checks	Complete
16	Create dynamic scaling policy	Complete
17	Select CPU utilization metric	Complete
18	Set threshold to 70%	Pending
19	Increase desired capacity by 2	Complete
20	CloudWatch monitors CPU	Complete

21	**Alarm triggers scaling policy**	**Test successful**
22	**Auto Scaling launches new instances**	**Complete**
23	**New instances install Apache through User Data**	**Configured**





Hello Parvez - Apache Auto Scaling

